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MetaShift-Bench: A Metadata-Anchored Audit Benchmark for PM2.5 Method-Transition Discontinuities

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MetaShift-Bench: A Metadata-Anchored Audit Benchmark for PM2.5 Method-Transition Discontinuities

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Abstract

Air-quality records are used for regulation, public communication, exposure assessment, and scientific analysis, yet their time series can display breaks for many reasons. An abrupt shift near a reported measurement-method transition may reflect a station-specific monitoring discontinuity, but it may also coincide with weather, smoke, emissions, data processing, or a broader regional movement. A single monitor cannot distinguish these explanations. This report introduces MetaShift-Bench, a reproducible audit benchmark for reported United States Environmental Protection Agency Air Quality System (AQS) PM2.5 Method Code transitions. A transition is treated as a pre-observation metadata anchor, never as proof of a physical replacement, instrument failure, or causal measurement bias. The protocol builds geographically constrained counterfactuals using distinct physical donor sites, tests cross-site and single-series methods under known synthetic perturbations, and retains every eligible real anchor with either diagnostics or an explicit machine-readable failure reason.

The frozen inventory contains 2,424,793 canonical daily observations from 1,689 monitor time series and 563 persistent anchors. Only 238 anchors have at least three distinct physical donors. Of all anchors, 228 receive the common cross-site comparison, while 325 are donor-insufficient and 10 have input-window failures. The stable-regime synthetic benchmark uses 146 physical target sites: 66 calibration sites select frozen decision thresholds, and 80 complete-input-disjoint sites supply held-out results. Standard synthetic control has local-effect MAE 0.09983, AUPRC 0.91476, macro-F1 0.81641, and regional false-positive rate 0.140. Cross-validated MetaShift has a lower MAE point estimate (0.09742) but lower AUPRC and macro-F1 than standard synthetic control; its paired MAE difference is -0.00240 with 95% bootstrap interval [-0.00772, 0.00347].

The paired interval crosses zero, so the frozen evidence does not support a general MetaShift superiority claim. On the same held-out design, fixed-weight conditional bootstrap intervals under-cover at nominal 95%, while split-conformal intervals over-cover at nominal 90%; real-event intervals are therefore diagnostic rather than calibrated physical-bias confidence intervals. The audit reports 34 supported candidates, 122 not-supported events, and 407 inconclusive events under predeclared observational rules. These are not instrument-fault labels. MetaShift-Bench's contribution is a complete, metadata-anchored evidence record that distinguishes an analyzable local residual pattern from unavailable evidence, preserves negative findings, and makes the limits of a cross-site audit reproducible.

Keywords: air-quality monitoring; PM2.5; metadata anchors; synthetic control; change-point attribution; reproducible audit.

Contents

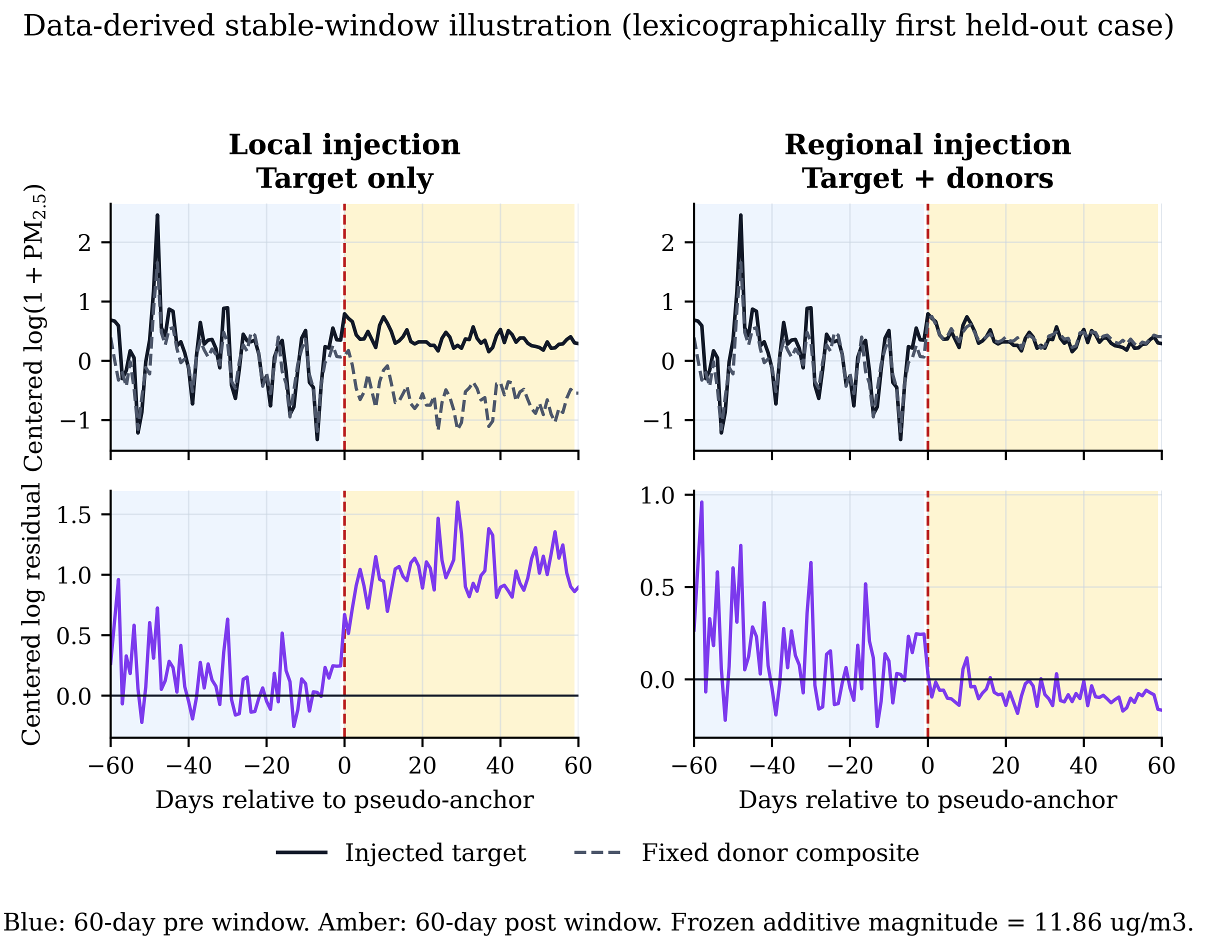

Figure 1: Data-derived stable-window illustration for the lexicographically first held-out case fixed in the display contract. The same frozen additive injection is applied to the target alone (left) or to the target and donor composite (right). Blue and amber bands identify the protocol-defined 60-day pre- and post-pseudo-anchor windows. The underlying baseline is checksum-pinned public data; the injected conditions are known synthetic truth, not a model or diagnosis of a real metadata anchor.

1 Introduction

Air-quality measurements support regulatory compliance, public communication, trend analysis, exposure assessment, and health research. A visible break in a monitor's PM2.5 series can have several explanations: meteorology, wildfire smoke, local emissions, siting, operations, data processing, or a change in reported measurement metadata. These explanations cannot be separated by a numerical change-point alone. The problem is consequently both scientific and computational: an analyst needs a reproducible way to join heterogeneous public records, construct time-respecting comparison sets, evaluate algorithms against known truth where possible, and record when the inputs do not justify an algorithmic answer.

Figure 1 illustrates the central distinction using a checksum-pinned held-out stable window rather than generic curves. At a pseudo-anchor, the same frozen additive magnitude is injected either into the target alone or into the target and its donor composite together. The first condition creates a known target-reference residual; the matched regional condition should not. This display is synthetic evaluation data derived from a stable observed window, not an observation or causal model of a real Method Code transition.

The AQS data model provides a narrow and auditable starting point. Its daily records expose a reported Method Code, Method Name, and Parameter Occurrence Code (POC) for each series

[26, 27, 30]. A persistent Method Code transition can therefore define a candidate date before any local residual has been inspected. It is not an instrument log: a POC is not a universal physical-instrument identifier, and a reported code transition may have several non-hardware explanations. This report therefore uses the term *metadata anchor*, rather than treatment, failure, or replacement.

MetaShift-Bench asks whether a target-minus-reference residual near a reported anchor is unusual relative to stable dates and reference sites. The intended output is an evidence audit, not an automatic correction of historical concentrations. Cross-site counterfactual methods can separate some station-specific and regional patterns, but their interpretation requires pre-event agreement, qualified reference series, and transparent uncertainty boundaries. Synthetic control and difference-in-differences provide useful counterfactual and placebo precedents [1, 2, 9]; they do not convert monitoring metadata into a randomized intervention.

The report addresses five predeclared questions:

1. **RQ1:** How do cross-site and single-series methods behave under known local and matched regional perturbations in stable monitoring windows?
2. **RQ2:** Does either frozen MetaShift variant establish a confidence-supported aggregate advantage over standard synthetic control?
3. **RQ3:** What fraction of the complete metadata-anchor universe supports a common cross-site comparison, and what do its placebo and sensitivity diagnostics show?
4. **RQ4:** How well do the reported interval constructions cover known synthetic effects, and what uncertainty remains uncalibrated?
5. **RQ5:** What contextual support is available from same-site POC records, official documents, and a separate PM2.5 parameter pipeline?

The contribution is a measurement-method transition *audit benchmark*, not a promoted estimator. More specifically, this report contributes:

1. a public-data provenance chain that preserves archive checksums, canonical cleaning rules, and a reproducible primary PM2.5 signal;
2. a metadata-anchor definition that is fixed before local-effect inspection and explicitly separates a monitor series from a physical site;
3. a distinct-physical-donor rule that prevents multiple POCs at one site from being counted as independent geographic evidence;
4. a stable-regime synthetic benchmark with target-only and matched regional perturbations, complete input-footprint split isolation, and frozen evaluation metrics;
5. a complete real-anchor ledger with counterfactual diagnostics, placebos, donor sensitivity, rule-based evidence tiers, and explicit abstention; and
6. a negative model-comparison result: neither frozen MetaShift variant establishes a confidence-supported aggregate advantage over standard synthetic control.

This framing is deliberately conservative. No taxonomy-stratified analysis is reported because the frozen Method Code taxonomy remains pending independent student and teacher review. The benchmark does not claim that a Method Code change proves a device replacement, a measurement failure, or a causal measurement bias. Instead, it supplies an auditable record of what the available public metadata and time series can support, what they cannot support, and which questions require records-based human investigation.

2 Problem formulation and claim boundaries

2.1 Objects, metadata anchors, and the target question

Let $s = (\text{state}, \text{county}, \text{site})$ denote a physical monitoring site, let p denote its reported POC, and let $y_{s,p,t}$ be the retained daily concentration at date t. A monitor *series* is the ordered pair (s, p); a physical site is s alone. This distinction is essential because different POCs can occur at the same location without furnishing independent geographic information. For the primary parameter, an anchor is

$$a = (s, p, t_0, m^-, m^+),$$

where t_0 is a persistent reported Method Code transition from m^- to m^+. The anchor is observable metadata, not a verified intervention.

For each anchor, the computational question is whether a pre-specified cross-site residual changes around t_0 more than would be expected from the benchmark's comparison procedures. It is *not* "did a device change?" or "what was the true pollution concentration?" These questions would require station histories, calibration records, deployment information, and possibly meteorological and emissions evidence that do not follow from an AQS Method Code field alone.

2.2 Time-respecting residual estimand

For a target series $i = (s, p)$ and qualified donor series $j \in N_a$, the analysis uses the fixed transformation

$$z_{i,t} = \log(1 + \max(y_{i,t}, 0)).$$

With nonnegative donor weights w_{ij} that sum to one, define the calibration-centered residual

$$r_{i,t} = z_{i,t} - \sum_{j \in N_a} w_{ij} z_{j,t} - \text{median}_{u \in T_{\text{cal}}} \left(z_{i,u} - \sum_{j \in N_a} w_{ij} z_{j,u} \right).$$

The implemented inclusive calibration slice runs from $t_0 - 180$ through $t_0 - 15$ (166 possible calendar dates), while the comparison slices run from $t_0 - 60$ through $t_0 - 1$ and from t_0 through $t_0 + 59$ (60 possible dates each). It requires at least 30 retained observations in each required window and at least two available donors on retained dates. The calibration and pre-anchor slices overlap on 46 calendar dates. No post-anchor outcome enters weight fitting or residual centering, but the overlap means that displayed pre-window residuals are not an independent validation set. This is a frozen implementation limitation, not a post-hoc sensitivity adjustment. The displayed protocol-defined effect is

$$\widehat{\Delta}_a = \text{median}_{t \in T_{\text{post}}} r_{i,t} - \text{median}_{t \in T_{\text{pre}}} r_{i,t}.$$

It is a local residual discontinuity on the log scale. It is not an estimate of a causal physical bias or a corrected concentration.

2.3 Claim boundaries

The benchmark's core logic is comparable to weakly labelled audit problems: an observed administrative marker gives a reproducible condition to inspect, but not a ground-truth class [12, 21]. Accordingly, the protocol makes three commitments. First, it fixes the anchor and the time-respecting donor construction before the local residual is measured. Second, it uses synthetic perturbations in stable regimes whenever an outcome requires known truth. Third, it preserves non-complete cases as abstentions rather than treating missing evidence as a negative result.

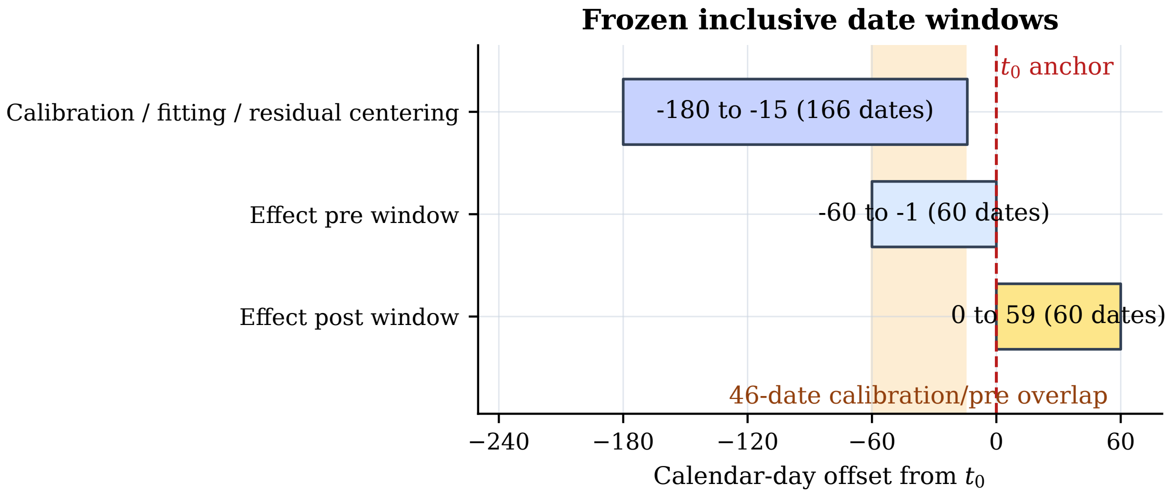

Figure 2: Inclusive date-window implementation used by the frozen primary real-anchor audit. The amber overlap is 46 calendar dates shared by fitting and the displayed pre-anchor comparison. No post-anchor outcomes are used for fitting, but the overlap prevents treating pre-window residuals as an independent validation set. The figure is an implementation audit, not a new estimator or re-analysis.

Table 1: Claim boundaries imposed by the MetaShift-Bench protocol.

The evidence can support	The evidence does not establish	Required interpretation
A reproducible metadata-anchored local residual audit	That a Method Code transition is a physical replacement or fault	Treat the transition as an anchor, not physical ground truth.
Benchmark-specific synthetic trade-offs among frozen methods	A robust aggregate MetaShift superiority claim	Report the paired intervals and retain the negative result.
Protocol-defined evidence tiers and abstentions	Instrument bias, true pollution correction, or causal attribution	Use tiers only to prioritize further records-based review.
Conditional interval behavior on frozen synthetic data	Coverage-calibrated confidence intervals for real-event physical bias	Describe real-event intervals as diagnostic.

Note. These boundaries are design constraints, not post hoc caveats.

2.4 What a negative model result means

MetaShift is evaluated as a metadata-informed weighting hypothesis alongside standard synthetic control, nearest-neighbor difference-in-differences, and single-series change-point baselines. The benchmark permits a benchmark-specific performance comparison, including paired uncertainty for local-effect error. It does not permit a conclusion of robust superiority from a favorable point estimate whose paired uncertainty interval crosses zero. This boundary protects against selection after inspecting the held-out partition and keeps the paper's main contribution focused on auditable benchmark construction.

3 Background and related work

3.1 Monitoring comparability and public metadata

Monitoring comparability is an empirical property rather than a label inferred from one administrative field. EPA publishes AQS file formats, Method Code and POC references, and continuous-monitor comparability resources [26, 27, 29, 30]. Collocated Federal Reference Method and Federal Equivalent Method studies show why configuration-specific comparison matters [17]. They do not, however, provide dated physical-change ground truth for every historical AQS Method Code transition.

The climate-record homogenization literature offers a useful structural analogy: pairwise comparisons and station histories can expose inhomogeneities, while the cause remains conditional on the available documentation [22]. Air-pollution analyses similarly use meteorological normalization to distinguish interventions or trends from changing weather [14, 16, 33]. Low-cost PM2.5 sensor studies further motivate calibration and reference comparison [5, 10, 11]. Their instruments, target populations, and calibration goals differ from a retrospective audit of regulatory-monitor metadata. MetaShift-Bench uses this literature as measurement-quality context, not as evidence that a particular AQS transition has a particular physical cause.

3.2 Reference series, counterfactuals, and change points

Single-series change-point procedures include cumulative-sum monitoring, segmentation methods such as PELT, and multiscale approaches [13, 18, 24]. These methods are valuable baselines because they find patterns in one sequence; they cannot by themselves determine whether a break is monitor-local or shared across a region. The difference motivates matched regional variants in this benchmark.

Synthetic control provides donor-weighted counterfactuals and in-space placebo logic [1, 2]. Generalized, augmented, and synthetic difference-in-differences extensions address other panel structures and regularization choices [3, 6, 32]. Modern difference-in-differences work also emphasizes assumptions and heterogeneity under varied treatment timing [9, 15]. MetaShift-Bench adopts the computational ideas of pre-event fitting and reference comparison, but does not claim that the causal identification assumptions of a policy study apply to a reported monitoring-method transition.

3.3 Uncertainty, selection, and abstention

Time-series resampling must respect serial dependence; circular moving-block bootstrap methods are a natural descriptive tool for this purpose [19]. Distribution-free predictive and conformal methods offer another way to assess interval behavior when exchangeability assumptions are

Table 2: Relationship between MetaShift-Bench and contributing research traditions.

Research tradition	Typical object and evidence	MetaShift-Bench boundary
Synthetic control and DiD [1, 3]	Pre-intervention panel outcomes and comparison units for an estimand under explicit assumptions.	Uses pre-anchor donor fitting and placebos as diagnostics; does not assert policy-style causal identification for metadata transitions.
Single-series change points [18, 24]	One sequence and a change statistic or estimated break.	Retains these methods as baselines but separates a target-local pattern from a matched regional movement only when a qualified reference is available.
Monitoring comparability and collocation [17, 29]	Configuration-specific technical comparisons, collocation, and station documentation.	Treats auxiliary POC, QA, and document records as contextual evidence; no metadata field is promoted to physical-device ground truth.
Climate-record homogenization [22]	Station histories plus pairwise evidence for record inhomogeneities.	Adopts the audit logic of preserving candidate discontinuities and requiring records for mechanism; it does not claim a homogenized corrected PM2.5 series.

made explicit [4, 20, 23, 31]. In this report, these methods motivate synthetic coverage evaluation rather than a claim that real monitoring events enjoy generic calibration.

The evidence-tier task is also related to multiple testing, selection, and abstention. Benjamini–Hochberg adjustment organizes an exploratory family of screening quantities [7]; it does not turn those quantities into causal probabilities. Post-selection inference highlights why selection can invalidate naive uncertainty statements [8]. Research on noisy labels underscores that observed proxies are not ground truth [12, 21]. MetaShift-Bench consequently records an inconclusive tier when required comparative evidence is unavailable instead of forcing a positive or negative physical diagnosis.

3.4 Contribution boundary

Table 2 locates the benchmark relative to its component literatures. The comparison concerns the scope of the reported workflow, not a claim that MetaShift-Bench improves the underlying methods.

MetaShift-Bench integrates established ingredients into one constrained workflow: AQS Method Code metadata defines anchors; distinct physical sites define geographic donors; stable windows receive known synthetic perturbations; and every eligible real anchor receives either a result or an explicit exclusion. We are not aware of a prior benchmark that jointly uses reported AQS measurement-method transitions as reproducible metadata anchors, cross-site diagnostic controls, stable-window synthetic perturbations, and complete eligible-event accounting. This is a scoped literature statement, not an assertion of being first in a broader field.

Table 3: Frozen v0.3.2 data and audit inventory.

Quantity	Count
Canonical daily PM2.5 records	2,424,793
Monitor time series	1,689
Stable pseudo-anchors from Method Code regimes	124
Stable pseudo-anchors from all-monitor regimes	22
Persistent Method Code anchors	563
Anchors with at least one distinct physical donor	394
Anchors with at least three distinct physical donors	238
Complete common-method comparisons	228
Donor-insufficient anchors	325
Estimator input-window failures	10

Note. The physical donor identifier is State Code + County Code + Site Number; at most one POC is retained per donor site.

4 Data and benchmark construction

4.1 Primary public-data corpus

The primary corpus comprises the seven annual EPA AirData daily archives for parameter 88101, PM2.5 local conditions, spanning 2019–2025 [26, 28]. The data gate records the official download URL, archive bytes, archive SHA-256, selected CSV member, retained CSV row count, and acquisition-time modification field for every archive. This provenance record permits reconstruction without versioning the large raw archives or any API response in the repository.

The canonical signal is the reported `Arithmetic Mean` for `24-HR BLK AVG` observations. The predeclared cleaning rule keeps finite values with observation percent at least 75, a non-excluded Event Type, and a nonempty Method Code. It then refuses duplicate records with the same state, county, site, POC, and date key. Thus alternative daily summaries cannot be treated as independent observations. The retained primary inventory is a frozen snapshot; it is not a claim that the public source is immutable or complete for all later AQS updates.

The daily data are public, but raw archives and API responses are not included in this manuscript repository or its paper assets. The display-only case-study renderer verifies archive checksums locally before reading the same source files. That separation supplies a reproducible source chain without placing raw data, credentials, or local cache files under version control.

4.2 Metadata anchors and distinct physical donors

An anchor is a persistent reported Method Code transition with at least 60 calendar days on each side, at least 45 retained observations in each adjacent method run, and no more than a seven-day transition gap. Its fields preserve the old and new reported codes and method names, but neither field establishes why the record changed. An eligible geographic donor must be at a different physical site, within the predeclared 100-km radius, method-stable over the target window, observed on at least 60 paired pre-event days, and correlated at least 0.60 on the log scale before the anchor.

Physical donor identity is State Code + County Code + Site Number. A target's candidate donor list can contain multiple POCs for one physical site, but the ranked donor rule retains at most one: highest pre-event correlation, then shorter distance, then more paired days, then POC as a deterministic tie-breaker. This rule remediates an earlier superseded inventory in which multiple POCs could be treated as independent geographic donors. Same-site alternate POCs are excluded from geographic donor construction and reserved for contextual consistency analysis.

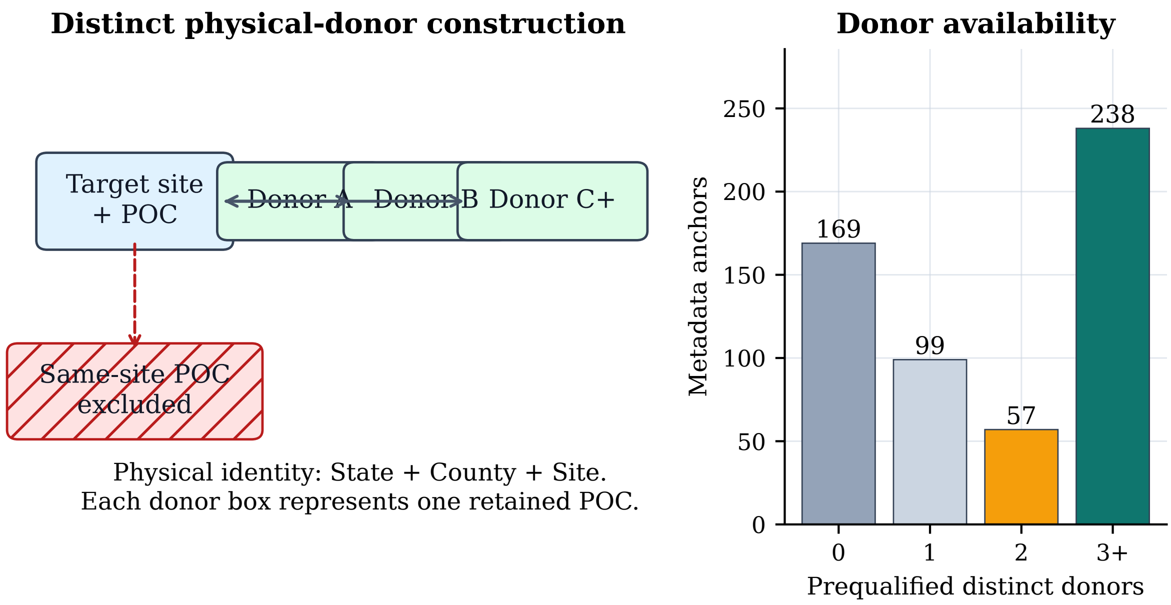

Figure 3: Distinct physical-donor construction and availability in the frozen 88101 inventory. A same-site alternate POC is explicitly excluded from the geographic donor set, and at most one ranked POC is retained at each other physical site. The right panel counts prequalified distinct donors before the common-method input-window check. These are audit availability counts, not counts of physical monitoring changes.

4.3 Stable-regime synthetic cases

Real anchors do not provide known physical-bias truth. The benchmark therefore constructs pseudo-anchors only inside stable target and donor method regimes, at least 60 days away from a reported target or donor transition. The frozen case manifest contains 124 method-transition stable regimes and 22 all-monitor stable regimes. Both sources are retained because the target is a known, stable measurement record at the pseudo-anchor; neither is used as a real-event causal label. Section 6 explains how the target-plus-donor footprints are split before any held-out metric is calculated.

4.4 Audit disposition and parameter separation

Every anchor is retained in the event audit. A complete comparison means that the required common cross-site inputs exist; donor insufficiency and input-window failure are explicit machine-readable outcomes rather than omitted rows. This complete accounting is necessary because a result on only the well-matched subset would overstate the scope of cross-site inference.

Parameter 88502 is processed through a separate data gate, anchor inventory, and audit table. It is neither appended to nor pooled with the primary 88101 results. This separation prevents an apparent replication from being created by mixing parameter-specific reporting conventions, availability, or eligible event populations.

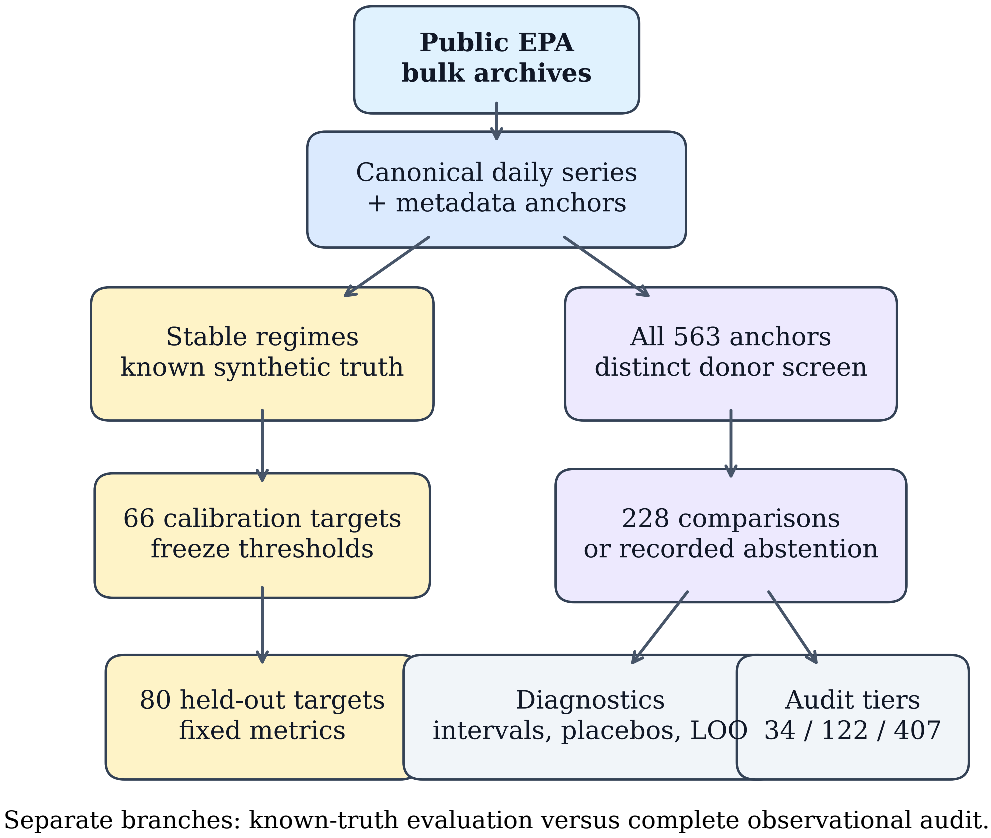

Figure 4: MetaShift-Bench evidence workflow. Public archives are canonicalized before persistent metadata anchors and distinct physical donor candidates are constructed. Stable regimes supply known synthetic truth, while every real anchor receives a common comparison or a recorded failure reason. The diagram is a workflow schematic; its arrows do not assert that metadata reveals a physical instrument state.

5 MetaShift-Bench audit framework

5.1 Counterfactual construction

For a complete anchor, the donor pool is ranked by pre-event correlation, distance, paired days, and POC only after physical-site deduplication. The nearest-neighbor difference-in-differences baseline places unit weight on the top-ranked donor. Standard synthetic control solves the nonnegative simplex problem

$$\min_{\substack{w_j \geq 0 \\ \sum_j w_j = 1}} \frac{1}{|T_{\mathrm{cal}}|} \sum_{t \in T_{\mathrm{cal}}} \left(z_{i,t} - \sum_{j \in N_a} w_j z_{j,t} \right)^2 .$$

The intercept is represented by calibration centering in the residual definition of Section 2. No post-anchor outcome is supplied to either method.

Fixed-weight MetaShift with a reliability-prior penalty starts from the pre-event reliability preference

$$\pi_{ij} \propto \max(\rho_{ij}, 0)^2 \exp(-d_{ij}/50),$$

where ρ_{ij} is the pre-event log-scale correlation and d_{ij} is distance in km. Its frozen real-audit objective is

$$\min_{\substack{w_j \geq 0 \\ \sum_j w_j = 1}} \mathrm{MSE}_{T_{\mathrm{cal}}}(z_i, Z_N w) + 0.1 \sum_j w_j^2 + 0.1 \sum_j (w_j - \pi_{ij})^2.$$

This is a reproducible regularized weighting rule, not a learned instrument taxonomy. In the synthetic benchmark only, cross-validated MetaShift tests a frozen set of penalty pairs on the final 45 pre-event days, chooses the smallest validation RMSE, and refits the selected pair on the supplied pre-event calibration table. It never sees anchor-post data during selection.

5.2 Two-level donor rule, algorithm, and computational scope

The audit uses two intentionally different donor conditions. An anchor is eligible for a common cross-site comparison only when at least three *distinct physical sites* pass the frozen geographic, stability, paired pre-event, and correlation screens. This protects against treating multiple POCs at a single location as independent geographic evidence. After dates are retained, a complete numerical comparison requires at least two available donors on a date and at least 30 retained observations in every required window. The latter condition protects calculation feasibility; it does not weaken the three-physical-donor eligibility rule.

For each anchor, the implementation follows the fixed sequence below:

1. identify and physically deduplicate prequalified donor candidates;
2. retain one deterministically ranked POC per donor site and abstain if fewer than three sites remain;
3. fit fixed cross-site weights using only the inclusive calibration slice, then center residuals on that same slice;
4. verify retained-date availability in calibration, pre, and post windows, recording an input-window failure when it is insufficient; and
5. compute the protocol-defined effect and diagnostic outputs only for a complete comparison.

If D candidates have T_{pre} paired pre-event records, screening and correlation assembly scale with the available paired data (on the order of DT_{pre}), and deterministic ranking adds $O(D \log D)$. The retained donor set is capped at three to five sites; residual assembly therefore scales linearly

Table 4: Predeclared primary evidence-tier rules and abstention outcomes.

Required diagnostic or condition	Threshold	Audit outcome
Complete common-method comparison	required	Missing input ⇒ inconclusive
Selection-aware nested interval	required	Unavailable ⇒ inconclusive
Unique stable post-transition time placebos	50	Fewer dates ⇒ inconclusive
Raw time-placebo probability	0.10	Above cutoff ⇒ not supported
BH-adjusted time-placebo q value	0.10	Unavailable ⇒ inconclusive; above cutoff ⇒ not supported
Fixed-weight conditional diagnostic interval	exclude 0	Otherwise ⇒ not supported
Leave-one-donor-out direction fraction	0.90	Below cutoff ⇒ not supported
All available required conditions	pass	Supported candidate discontinuity

Note. The rules synthesize observational diagnostics; they are not a supervised classifier, instrument-fault label, or physical-causality test. The nested interval is required for tier assignment but is not claimed to have synthetic coverage calibration.

in the retained window length and this small set size. The constrained numerical fit has solver-dependent iteration cost, so the report does not claim a hardware-independent closed-form runtime. The complete audit ledger reports failures rather than treating an optimization or input failure as a negative physical finding.

5.3 Single-series comparators

The benchmark includes five transparent single-series baselines: before-after median, Bayesian mean shift, CUSUM, rolling-MAD, and PELT. Before-after median and Bayesian mean shift estimate a known-date 60-day level change. CUSUM, rolling-MAD, and PELT score or locate changes in the single target series [13, 18, 24]. They serve a necessary contrast: a method can detect an abrupt time-series pattern without establishing that the pattern is local rather than regional.

5.4 Interval and falsification diagnostics

For fixed donor weights, the primary interval resamples pre- and post-anchor residuals using circular moving blocks of seven days, with 1,000 fixed-seed repetitions. It is conditional on previously chosen donors and weights; it does not include selection uncertainty. The selection-aware nested procedure re-block-resamples the calibration matrix, recomputes correlation eligibility, selects three to five donors from the fixed candidate pool, refits the nonnegative MetaShift weights, and separately resamples the pre/post residual windows. It preserves temporal ordering and reports failed reselection or effect replicates rather than silently replacing them.

The audit also computes stable post-transition time placebos, donor-as-treated placebos, a date-resampling comparison, and leave-one-donor-out refits. These procedures are falsification and robustness diagnostics. They do not create random assignment, prove a physical mechanism, or make a residual effect a measured instrument bias.

5.5 Rule-based evidence synthesis and abstention

The tier logic distinguishes unavailable evidence from failed available diagnostics. An event is inconclusive if it lacks a common comparison, selection-aware interval, sufficient time placebos, an adjusted placebo quantity, or donor sensitivity. It is not supported when those required inputs are available but a pre-event fit, fixed diagnostic interval, placebo screen, or donor-direction condition fails. A supported candidate satisfies all available required conditions. Benjamini–Hochberg adjustment is used only as an exploratory multiple-screening quantity [7]; the resulting tiers are

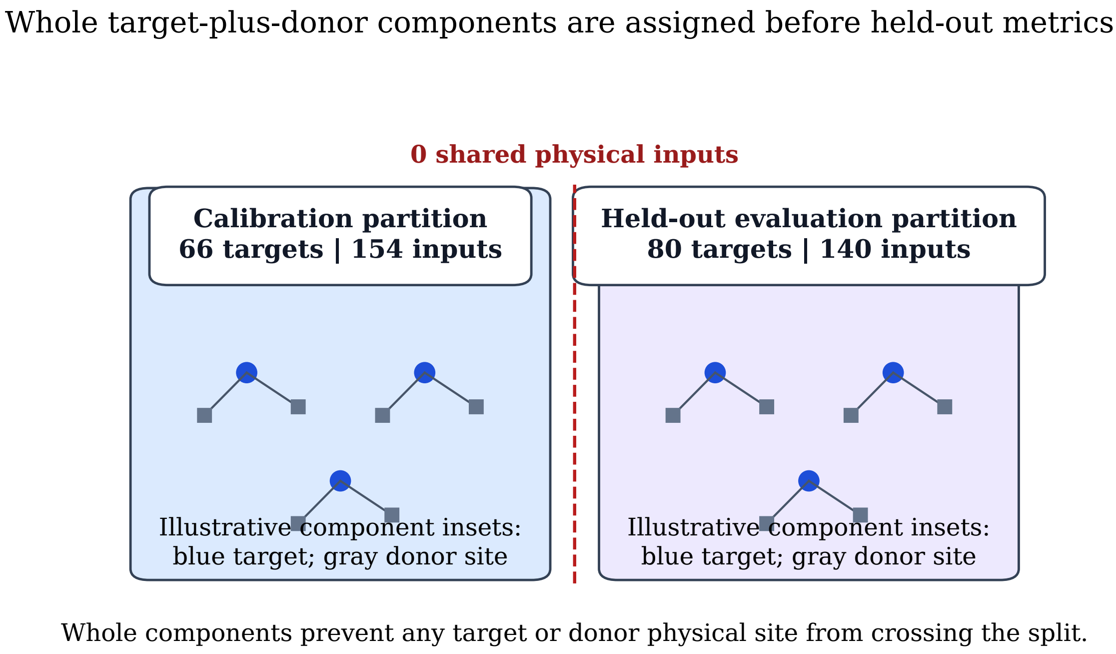

Figure 5: Leakage-resistant stable-case split, including the complete target-plus-donor physical input footprint rather than target sites alone. Whole connected components are assigned to calibration or held-out evaluation before metrics are computed; the frozen split audit records zero shared physical inputs. The small networks are explanatory insets, not a geographic map. This controls physical-input reuse across partitions; it does not make the synthetic perturbations a model of every real monitoring process.

observational audit summaries, not labels for instrument failure, replacement, bias, or pollution correction.

6 Experimental design

6.1 Stable-regime synthetic benchmark

Real metadata anchors have no physical-bias ground truth. Known-effect evaluation therefore uses only stable target and donor regimes separated from nearby reported transitions. The frozen `stable_full_v2` manifest contains 146 physical target sites. Whole connected components of the complete target-plus-donor physical input graph were assigned to the 66-site calibration subset or the 80-site held-out evaluation subset, so no physical input site crosses the split.

The benchmark injects target-only additive, proportional, gradual-drift, temporary-step, and variance changes, together with matched target-and-donor regional variants. Each perturbation variant contains 400 held-out evaluation samples. Each family therefore combines a local condition and a matched regional condition; the family rows in the results are not local-only effect estimates. Five frozen magnitude multipliers (0.5, 1.0, 1.5, 2.0, and 3.0) scale an additive step, proportional step, gradual drift, temporary step, or variance increase. The variance-only condition has no defined level-effect truth, so local-effect MAE is correctly reported as N/A for that condition.

The effect shape, magnitudes, seeds, candidate rule, model settings, and metrics were frozen before evaluation. For every method, a ranking threshold is selected only by macro-F1 on the calibration partition and then applied unchanged to the evaluation partition. The viewed evaluation partition is not used for model, threshold, donor-weight, quality-gate, or taxonomy adjustment. This is a protocol safeguard, not a claim that one finite split eliminates all future generalization uncertainty.

6.2 Benchmark metrics and uncertainty

For target-only non-variance perturbations, local-effect MAE is the mean absolute difference between the estimated and injected 60-day median log effect. AUPRC measures ranking of local against matched regional variants; macro-F1, precision, recall, and regional false-positive rate use the calibration-only threshold. CUSUM, PELT, and rolling-MAD have no protocol-defined effect-magnitude estimate, so their MAE cells are N/A while their ranking metrics remain observable.

The MetaShift-versus-standard-MAE comparison uses a paired cluster bootstrap over stable case IDs with 1,000 repetitions and fixed seed 20260830. Pairing keeps a method comparison tied to the same pseudo-anchor, perturbation, magnitude, and injected seed. It reports uncertainty for the frozen benchmark comparison, rather than a universal probability that one method dominates another.

6.3 Real-anchor audit and sensitivity

The real-anchor workflow uses the same distinct-physical-donor construction, pre-event fitting boundaries, and effect window across the three common cross-site methods. The five single-series baselines use the same 60-day before/after window but do not receive a geographic counterfactual. The audit records complete comparisons and explicit non-complete dispositions for all 563 anchors. Cross-validated MetaShift is evaluated only in the synthetic benchmark; it is not introduced into the real-audit comparison after observing those events.

Predeclared sensitivity analyses vary one factor at a time: donor-screening rules, 45/60/90-day effect windows, raw versus log reporting scale, and evidence-tier thresholds. The sensitivity rows reveal availability and interpretation dependence; they are not a search for the setting with the most favorable residual pattern. The standard synthetic-control rows in the main and reliability-ablation experiments are checked for exact alignment to tolerance 10^{-10}, preventing a seed or input mismatch from masquerading as an ablation finding.

6.4 Interval coverage protocol

Fixed-weight interval coverage is evaluated on held-out synthetic effects. The confidence level, seven-day block length, perturbation types, and 1,000 repetitions are fixed in the frozen configuration. The separate full selection-aware synthetic coverage protocol was frozen before outcomes. Its feasibility assessment records that full donor re-screening and weight refitting at the specified nested scale is infeasible within the submission deadline. The report therefore does not convert real-event selection-aware intervals into coverage-calibrated physical-bias intervals.

6.5 External context and reproducibility

Same-site alternate-POC records, narrow hourly POC windows, QA availability, and targeted official-document review supply contextual checks with known limitations. A separate parameter-88502 pipeline uses its own scan and audit; it is never pooled with 88101. The frozen release records the exact source commit, locked Python environment, source manifests, machine-readable checks, and a two-environment comparison of designated core artifact hashes.

Table 5: Independent stable-regime synthetic evaluation.

Method	MAE	AUPRC	Macro-F1	Regional FPR
Standard SC	0.09983	0.91476	0.81641	0.140
MetaShift fixed	0.10344	0.89832	0.79515	0.150
MetaShift CV	0.09742	0.90178	0.80916	0.135
Nearest-neighbor DiD	0.11150	0.88443	0.78852	0.075

Note. Metrics are aggregates over the frozen `stable_full_v2` evaluation set. Lower MAE and regional false-positive rate are preferable; higher AUPRC and macro-F1 are preferable.

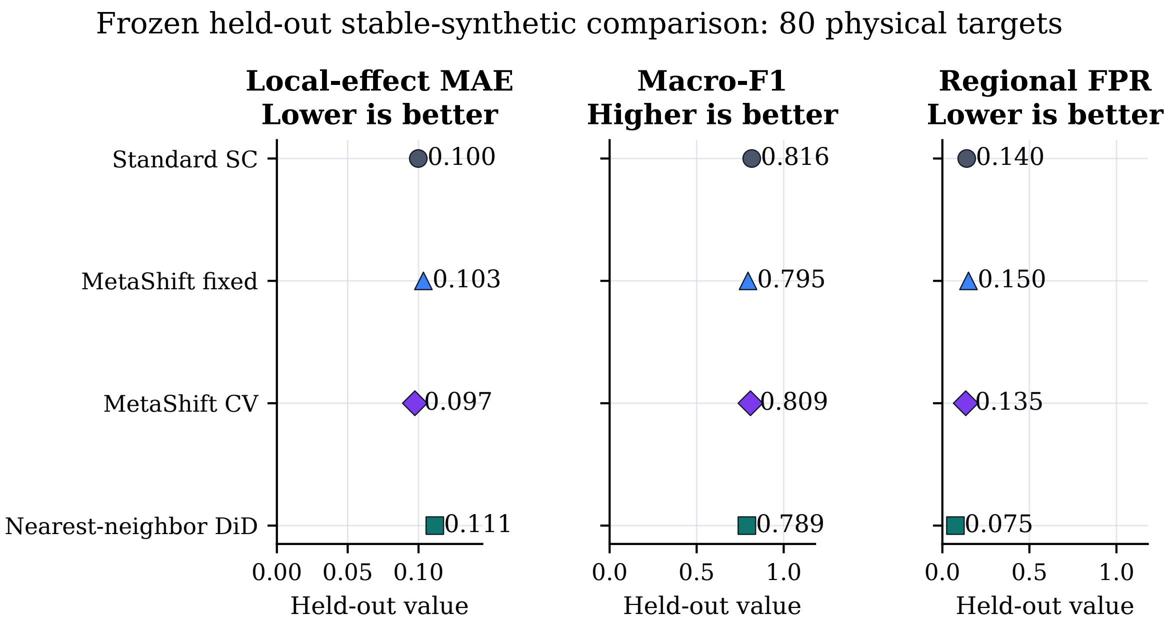

Figure 6: Held-out stable-regime synthetic comparison for the four cross-site methods. Dots show local-effect MAE, macro-F1, and regional false-positive rate on the frozen 80-site complete-input-disjoint partition; AUPRC is retained in Table 5. Lower MAE and regional false-positive rate are preferable, whereas higher macro-F1 is preferable. The panels do not form a post-hoc composite ranking; all values originate in the saved `stable_full_v2` metrics table.

7 Results

7.1 RQ1: Stable-window synthetic comparison

Answer. The frozen benchmark distinguishes cross-site methods from single-series baselines, but no method is uniformly best across all metrics and paired perturbation families. Table 5 reports the four cross-site methods used for the primary comparison, while Table 6 restores all nine frozen methods. Figure 6 visualizes the main cross-site trade-offs without collapsing them into a post-hoc composite score.

Standard synthetic control has stronger aggregate attribution ranking and macro-F1 than both MetaShift variants, while cross-validated MetaShift has the smallest aggregate local-effect MAE. Nearest-neighbor difference-in-differences has the lowest regional false-positive rate among these four methods but weaker effect error and attribution metrics. These are benchmark-specific trade-offs, not a ranking for every monitoring network.

The single-series methods remain in the complete comparison because an anchor date alone can make a time-series break appear compelling. Their inclusion shows what is lost when a

Table 6: Complete frozen aggregate comparison of all benchmark methods.

Method	Local-effect MAE	AUPRC	Macro-F1	Regional FPR
Standard SC	0.09983	0.91476	0.81641	0.140
MetaShift fixed	0.10344	0.89832	0.79515	0.150
MetaShift CV	0.09742	0.90178	0.80916	0.135
Nearest-neighbor DiD	0.11150	0.88443	0.78852	0.075
Bayesian mean shift	0.24672	0.50116	0.49962	0.542
Before-after median	0.25330	0.50125	0.50024	0.505
CUSUM	N/A	0.50117	0.49720	0.581
Rolling MAD	N/A	0.50112	0.49427	0.402
PELT	N/A	0.50025	0.49976	0.522

Note. All values are aggregates over the held-out stable-regime evaluation partition. Lower MAE and regional FPR are preferable; higher AUPRC and macro-F1 are preferable. N/A means that the method supplies no local-effect magnitude estimate for that perturbation design, not that the method was omitted.

Table 7: Paired event-cluster bootstrap for local-effect MAE differences.

Comparison	Difference	95% bootstrap interval	Clusters
MetaShift fixed minus Standard SC	0.00361	[-0.00401, 0.01163]	80
MetaShift CV minus Standard SC	-0.00240	[-0.00772, 0.00347]	80

Note. Difference is MetaShift minus standard synthetic control; negative values favor MetaShift on MAE. Both bootstrap intervals include zero.

method does not ask whether target and donors moved together. N/A local-effect MAE cells are retained rather than replaced by a surrogate metric: CUSUM, PELT, and rolling-MAD supply change evidence but no protocol-defined level-effect estimator for the variance-only condition.

The perturbation matrix makes the scope of the aggregate result visible. Cross-site methods distinguish additive and gradual target-local changes much better than the single-series baselines under this design, whereas proportional, temporary, and variance conditions expose different false-positive and ranking trade-offs. Because each row aggregates a paired local and regional condition, it tests local-versus-regional discrimination rather than a generic detection-rate claim.

7.2 RQ2: Bound on a MetaShift superiority claim

Answer. Neither MetaShift variant has a confidence-supported aggregate MAE advantage over standard synthetic control. The paired bootstrap intervals cross zero for both frozen variants, including cross-validated MetaShift's -0.00240 difference with interval [-0.00772, 0.00347].

The main and ablation experiments share 7300 standard synthetic-control records exactly to the configured numerical tolerance. Table 8 therefore compares reliability-prior variants without changing the standard-control reference rows. The ablations do not establish a stable overall MetaShift advantage; this negative result closes the model-superiority route for the frozen benchmark.

7.3 RQ3: Complete real-anchor audit

Answer. Only 228 of 563 anchors support a complete common cross-site comparison under the distinct-physical-donor rule. The remaining 325 donor-insufficient and 10 input-window events are retained as applicability boundaries rather than removed.

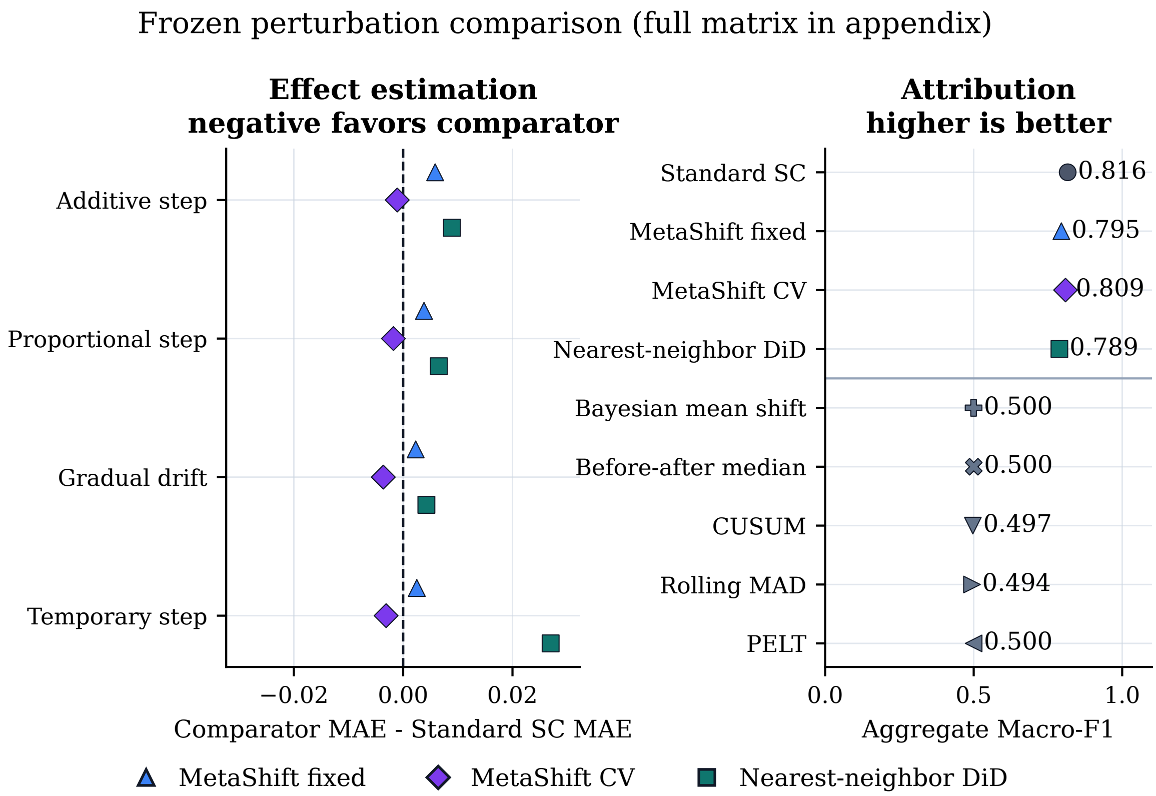

Figure 7: Main-text perturbation comparison. The left panel gives each non-variance perturbation family's local-effect MAE difference from standard synthetic control for the three cross-site comparators; negative values favor the comparator on that metric. The right panel retains aggregate macro-F1 for all nine frozen methods, separated by cross-site versus single-series status. The complete five-family, all-metric matrix remains in Table 17; variance-only local-effect MAE is N/A by definition, not an exclusion.

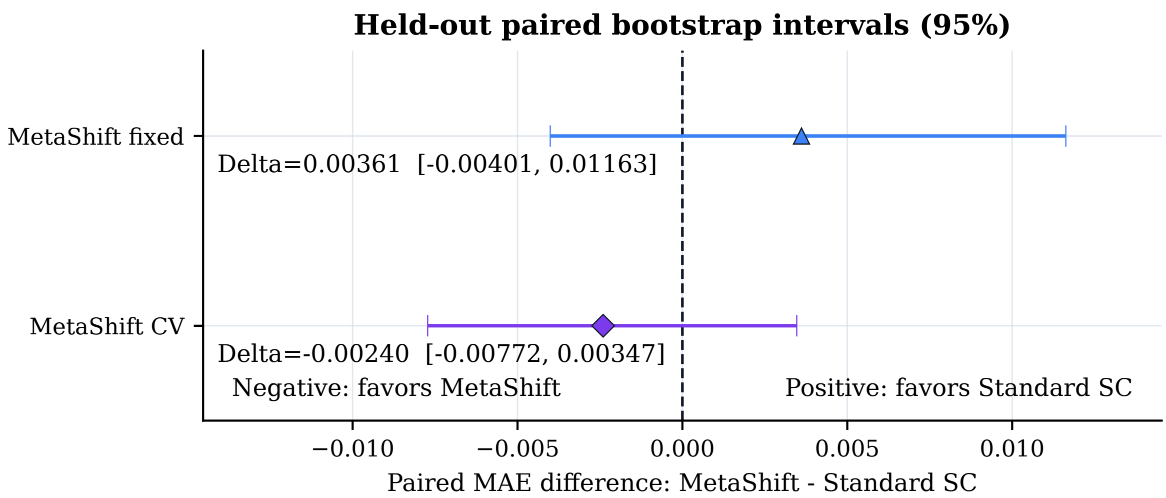

Figure 8: Paired event-cluster bootstrap intervals for MetaShift minus standard synthetic-control local-effect MAE on matched held-out synthetic instances. The dashed vertical line is no difference; negative values favor MetaShift on MAE. Both frozen intervals cross zero, so the figure does not support a confidence-supported aggregate superiority claim.

Table 8: Reliability-prior and regularization ablations on shared synthetic inputs.

Variant	MAE	Macro-F1	Regional FPR
Standard SC	0.09983	0.81641	0.140
No ridge penalty	0.10289	0.80127	0.138
Ridge = 0.01	0.10304	0.80263	0.130
Add coverage	0.10332	0.79862	0.135
MetaShift full prior	0.10344	0.79515	0.150
No reliability-prior penalty	0.10350	0.80167	0.133
No correlation term	0.10362	0.79858	0.133
No distance term	0.10370	0.80588	0.088
Uniform prior	0.10378	0.80521	0.112
Ridge = 1.0	0.10500	0.80285	0.097
Direct reliability	0.10543	0.80574	0.105

Note. The common standard-synthetic-control records align exactly across the main and ablation experiments to tolerance 10^{-10}.

Table 9: Complete 88101 event audit and real-event diagnostics.

Audit disposition	Anchors
Complete common-method comparison	228
Fewer than three distinct physical donors	325
Estimator input-window failure	10
Total	563

Method	Events	Fixed conditional interval excludes 0	Median log effect
MetaShift fixed	228	159	-0.08704
Standard SC	228	145	-0.07078
Nearest-neighbor DiD	228	134	-0.07490

Note. Nested selection-aware intervals are available for 227/228 complete events; 153 exclude zero. Their mean width is 0.19057 log units versus 0.17863 for fixed-weight MetaShift intervals. Leave-one-donor-out refits complete for 227 events, with 202 retaining direction under every donor removal. All intervals are diagnostic, not calibrated physical-bias confidence intervals.

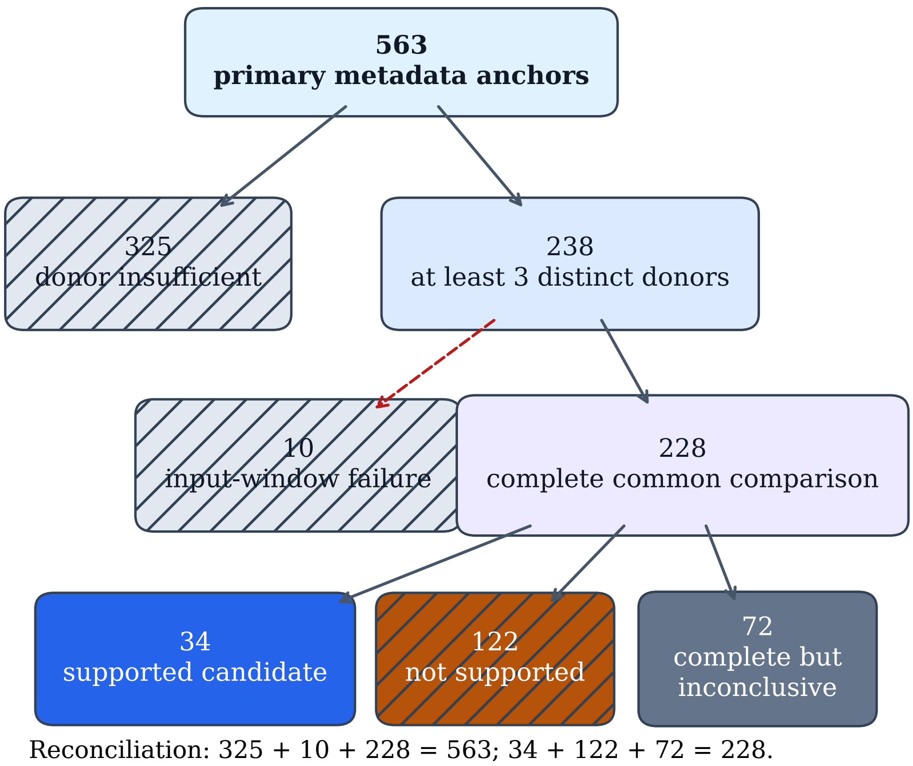

Figure 9: Hierarchical accounting of all 563 primary metadata anchors. Donor-insufficient and input-window-failure rows remain inconclusive audit dispositions. Only complete comparisons flow into the observational three-way tier rules; the lower reconciliation confirms both count partitions. No branch represents a physical monitoring diagnosis.

Table 10: Placebo, external-document, and same-site POC evidence boundaries.

Diagnostic	Value
Complete time-placebo events with at least 50 dates	157
Complete time-placebo events with 100 dates	128
Raw time-placebo probabilities at most 0.10	71
BH q values at most 0.10	40
Donor-as-treated placebo records	866
Date-resampling permutations	200
Dated site-specific external confirmations	0/20
Usable hourly same-site POC comparisons	9/11
Daily/hourly POC direction agreements	8/9
Standard SC 90th-percentile risk-gate retention	75/80

Note. Time and donor placebos are diagnostic falsifications. Same-site POC and document review provide contextual evidence only, not instrument ground truth.

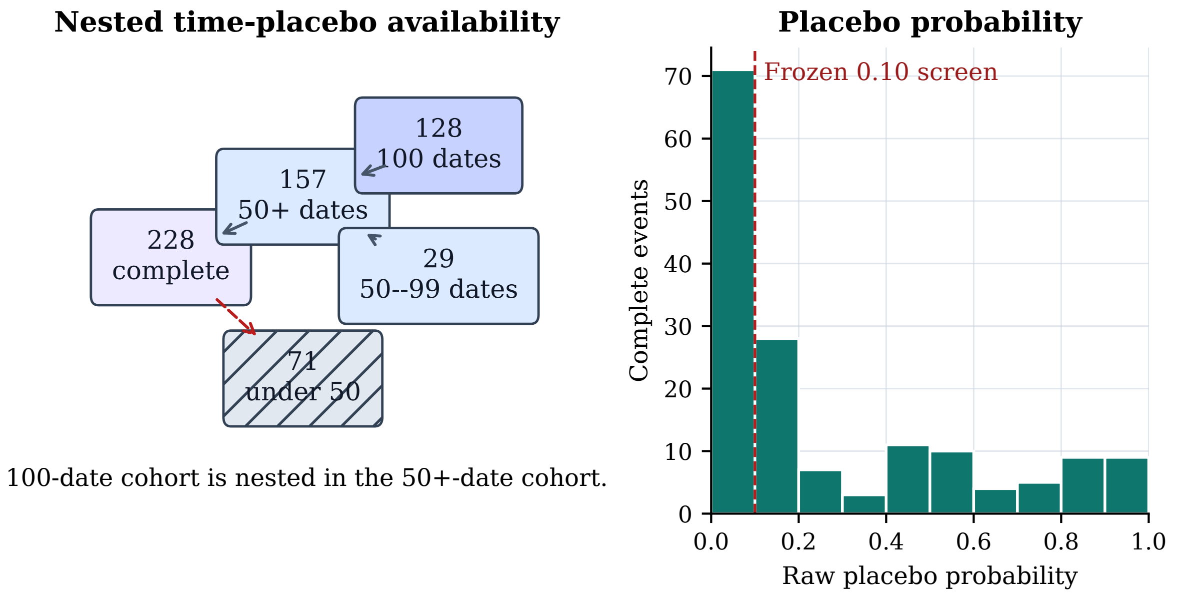

Figure 10: Time-placebo availability and raw within-event placebo-probability distribution. The vertical line marks the frozen exploratory 0.10 screening threshold. These values are diagnostic falsification quantities, not causal probabilities; incomplete placebo sets remain visible in the left panel.

The exploratory evidence synthesis assigns 34 supported candidates, 122 not-supported events, and 407 inconclusive events. The supported label means that predeclared diagnostic conditions were jointly met; it does not confirm a physical failure, replacement, or causal bias.

Time-placebo coverage is available for 157 complete events with at least 50 dates, including 128 with 100 dates. The donor-as-treated table contains 866 rows, and the 200-resample date comparison has upper-tail probability 0.00498. These are diagnostic falsifications rather than causal probabilities.

7.4 RQ4: Interval coverage and sensitivity

Answer. The two available fixed-protocol interval constructions do not support a generic calibration claim. At nominal 95%, fixed-weight conditional block-bootstrap coverage ranges from 63.875% to 67.281% over 3,200 held-out effect instances per method. At nominal 90%, split-conformal coverage ranges from 98.8125% to 99.5625% and is highly conservative.

Selection-aware nested intervals exist for 227 complete real events and exclude zero for 153.

Table 11: Held-out synthetic interval coverage by method.

Method	Fixed 95% coverage	Fixed mean width	Conformal 90% coverage	Conformal mean width
Standard SC	67.281%	0.265	98.8125%	0.396
MetaShift fixed	63.875%	0.270	99.4375%	0.471
MetaShift CV	65.406%	0.265	98.9375%	0.437
Nearest-neighbor DiD	66.656%	0.287	99.5625%	0.565

Note. Each method has 3,200 evaluation effect instances. Fixed-weight intervals under-cover and split-conformal intervals over-cover under this frozen synthetic design.

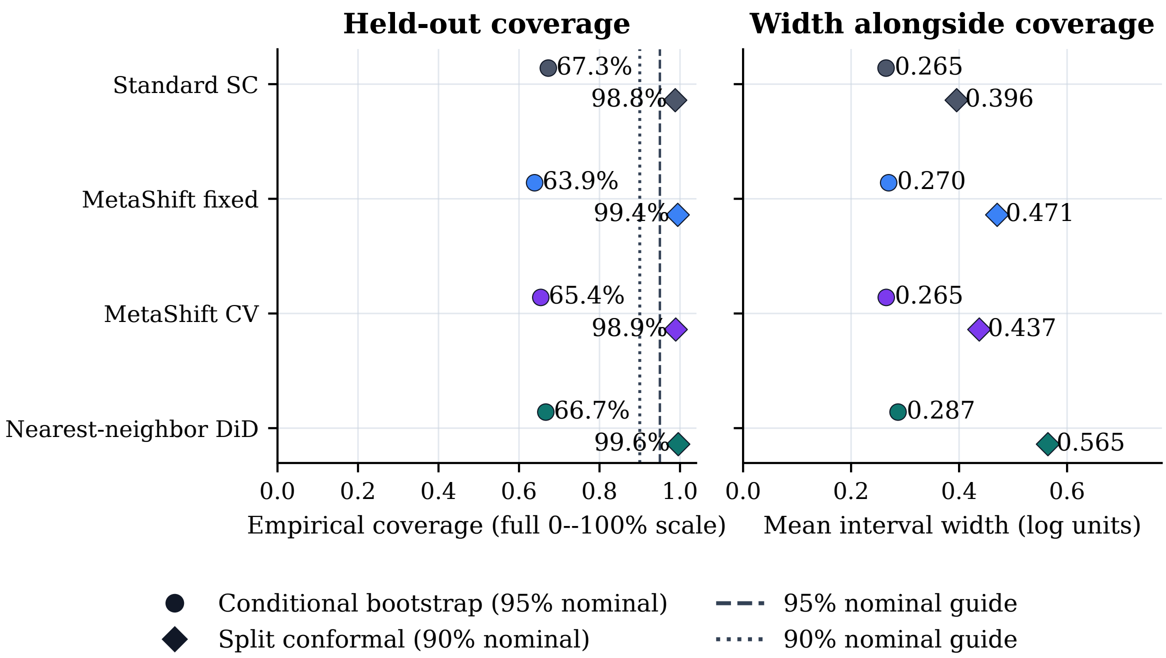

Figure 11: Empirical held-out synthetic coverage and mean interval width for four cross-site methods. Circles are fixed-weight conditional moving-block intervals at nominal 95%; diamonds are split-conformal intervals at nominal 90%. The coverage panel intentionally uses the full 0–100% scale, and paired width values show the cost of conservative coverage. Dashed reference lines identify nominal targets. Undercoverage and overcoverage are negative calibration findings, not parameters tuned after inspection.

Table 12: Predeclared effect-window and reporting-scale sensitivity.

MetaShift window	Complete events	Median log effect	Sign agreement with 60 days
45 days	224	-0.08769	93.3%
60 days	228	-0.08704	reference
90 days	196	-0.08064	93.9%

Method	Log/raw sign agreement	Spearman ρ
MetaShift fixed	95.2%	0.829
Standard SC	92.5%	0.868
Nearest-neighbor DiD	92.1%	0.843

Note. Window sensitivity adds a full method-stability check for every target and donor window. Reporting-scale agreement is a concordance diagnostic and does not equate raw and log causal estimands.

Table 13: One-factor screening sensitivity with three required distinct donors.

Setting	Eligible before donor rule	With at least 3 donors
Primary (100 km)	563	238
Coverage 70%	563	243
Coverage 80%	563	232
Stable window 45 days	572	252
Stable window 90 days	543	214
Transition gap 3 days	512	212
Transition gap 14 days	589	254
Donor radius 50 km	563	102
Donor radius 200 km	563	426
Correlation $\rho \geq 0.50$	563	241
Correlation $\rho \geq 0.70$	563	230

Note. One predeclared setting changes at a time. These counts describe audit availability, not detected-effect frequency.

Each requested 1,000 repetitions and retained at least 521 valid repetitions. This result does not validate their coverage: the frozen full selection-aware synthetic coverage protocol is infeasible within the deadline. The intervals are therefore reported as conditional diagnostics.

Table 12 reports predeclared effect-window and reporting-scale sensitivity; Table 13 reports one-factor screening availability. The geographic donor radius visibly changes how many anchors can be analyzed, so these settings are documented rather than optimized after viewing results.

7.5 RQ5: External context and transfer boundary

Answer. External evidence supplies limited contextual support, not ground truth. Of 11 same-site POC candidates, 9 have usable paired hourly windows and 8 agree in daily/hourly direction. The targeted official-document review found 0/20 dated site-specific confirmations. A missing public notice is unavailable evidence, not evidence that no physical change occurred.

The separately pinned QA-collocation review contains 12 candidate records, of which 2 match the target POC and 0 provide adequate matched pre/post windows. It is therefore an evidence-availability result, not a validation of a measurement-bias hypothesis.

The independent 88502 pipeline has 34 eligible metadata anchors and only 3 complete common-method comparisons. It demonstrates separate-pipeline feasibility but is too sparse to support broad generalization. EPA documentation about network data alignment provides useful method context [25]; it is not a dated site-specific confirmation for the reviewed anchors.

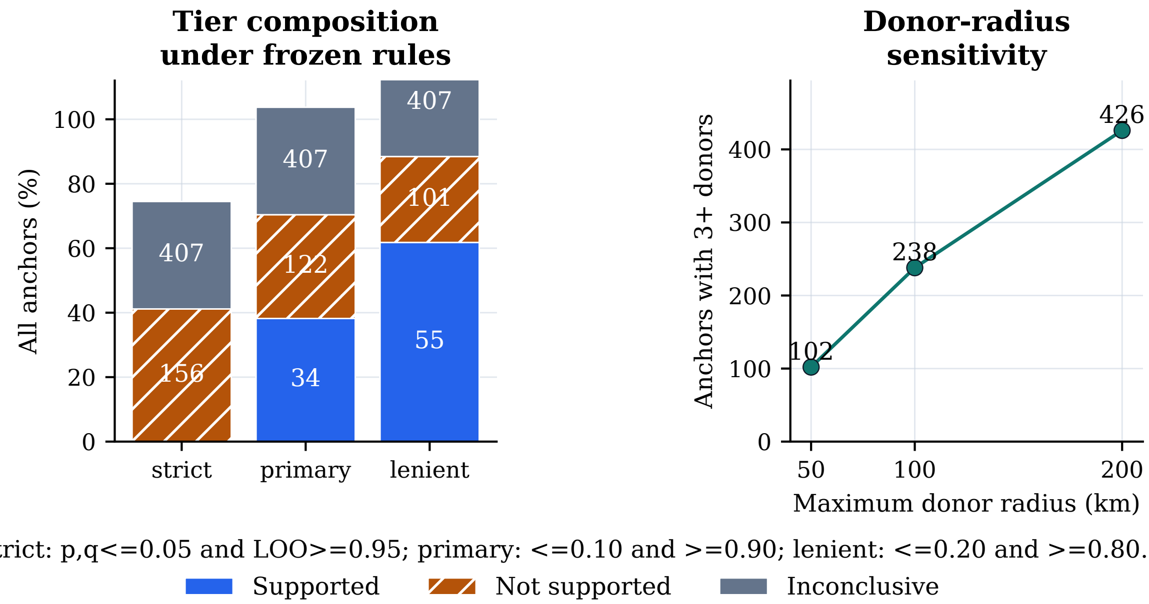

Figure 12: Predeclared evidence-tier and geographic-screening sensitivity. The left panel normalizes tier composition to all anchors while retaining counts in the bars; the right panel shows how the three-donor availability boundary changes with the predeclared radius. These describe rule dependence and data availability, not physical causality or a search for a favorable specification.

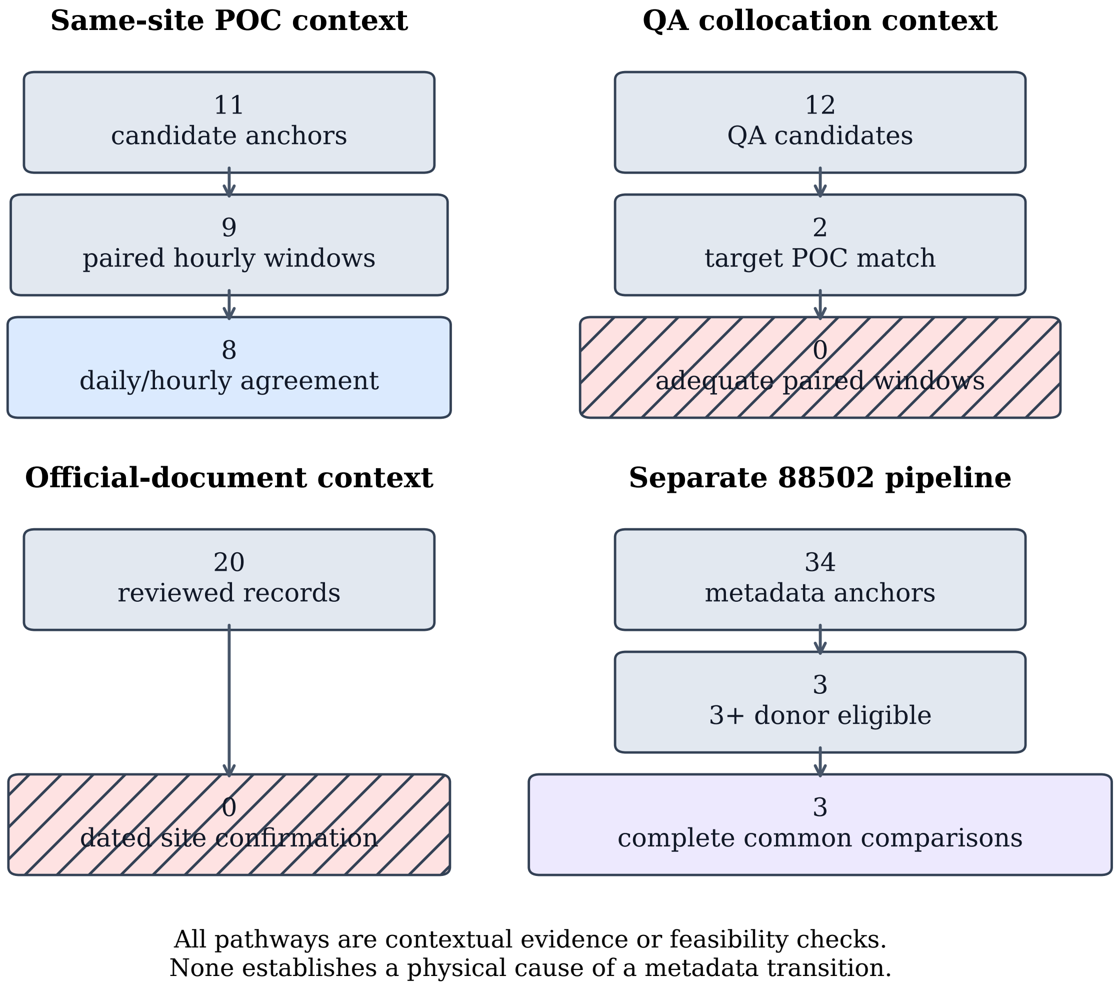

Figure 13: Four contextual-evidence ladders: same-site POC, QA collocation, targeted official-document review, and the separate 88502 pipeline. Each ladder visibly retains unavailable evidence, including zero adequate QA matched pre/post windows and zero dated site-specific confirmations. Neither a POC agreement nor an unlocated document supplies universal physical-instrument ground truth.

8 Representative case studies

8.1 Selection policy and source provenance

Examples can make an audit readable, but selected examples can also conceal a failure mode or overstate a method's scope. The three cases in this section are therefore not hand-picked for their visual contrast or effect magnitude. The display configuration selects one complete supported-candidate anchor and one complete not-supported anchor by minimum distance to the within-tier median standardized score, with anchor-ID tie-breaking. It selects the lexicographically first donor-insufficient inconclusive anchor to make the absence of a qualified counterfactual visible. The configuration, selected donors, archive manifest checksum, donor-table checksum, fitted weights, and reconstruction error are stored in the generated case-study manifest.

For the two complete cases, the renderer reads only the checksum-pinned public EPA archives, applies the frozen cleaning rule, reconstructs fixed-weight MetaShift with the reliability-prior penalty from pre-anchor data, and fails if the reconstructed saved log effect differs by more than 10^{-9}. The raw archives remain outside version control. The third case intentionally has no reconstructed counterfactual: its audit disposition is that fewer than three qualified distinct physical donors were available.

8.2 Interpretation boundaries

The supported-candidate panel illustrates a case in which the predeclared diagnostic conditions jointly hold. That status prioritizes the anchor for further station-history and technical-record review; it does not identify a device, a maintenance action, a calibration change, or a measurement bias. The not-supported panel illustrates the opposite distinction: a visible target series and a complete counterfactual are not sufficient when available diagnostics fail the frozen evidence rules. It would be misleading to discard that case merely because its residual plot is less dramatic.

The inconclusive panel is equally important. A target series may have an apparent before/after pattern, yet the protocol cannot form the common cross-site comparison if the distinct-donor requirement fails. MetaShift-Bench therefore abstains rather than borrowing a same-site POC, relaxing the donor rule after inspection, or assigning a physical explanation. This case makes the audit's coverage boundary concrete: lack of a qualified comparison is lack of evidence for this protocol, not evidence that nothing happened.

Table 14: Deterministically selected representative audit cases.

Case	Metadata anchor	Disposition	Saved diagnostic summary
Supported candidate	`12-057-0113 poc1-2023-12-01`	Complete	Effect -0.12777; fixed [-0.17258, -0.10511]; nested [-0.17399, -0.10560]
Not supported	`06-031-0004 poc8-2021-01-12`	Complete	Effect -0.08240; fixed [-0.21609, 0.03012]; nested [-0.24309, 0.03662]
Inconclusive	`01-103-0011 poc3-2023-02-01`	Donor insufficient	No counterfactual, effect, or interval is imputed.

Note. Supported and not-supported cases minimize absolute distance to their within-tier median standardized score, with anchor ID tie-breaking. The inconclusive case is the lexicographically first donor-insufficient anchor and deliberately has no counterfactual or interval. Intervals are diagnostic, not calibrated physical-bias confidence intervals.

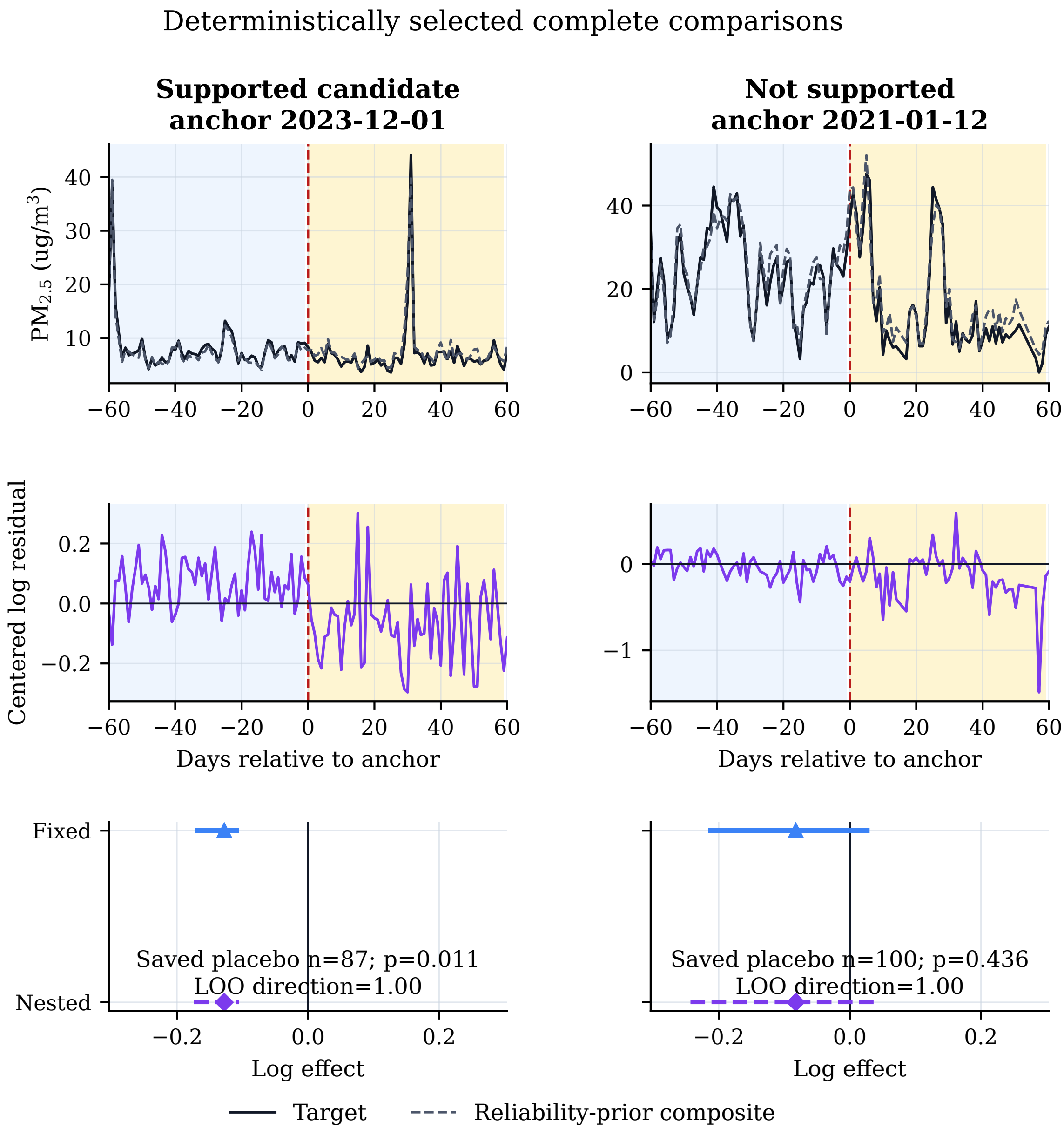

Figure 14a: Deterministically selected complete representative cases. The top row shows retained daily target values and the pre-anchor-fitted reliability-prior MetaShift counterfactual in $\mu g/m^3$; the middle row shows calibrated log residuals; and the lower row shows saved placebo summaries with fixed and nested diagnostic intervals. Blue and amber shading mark the 60-day pre- and post-anchor comparison windows. The figure displays saved diagnostic outputs, not evidence of a physical instrument event.

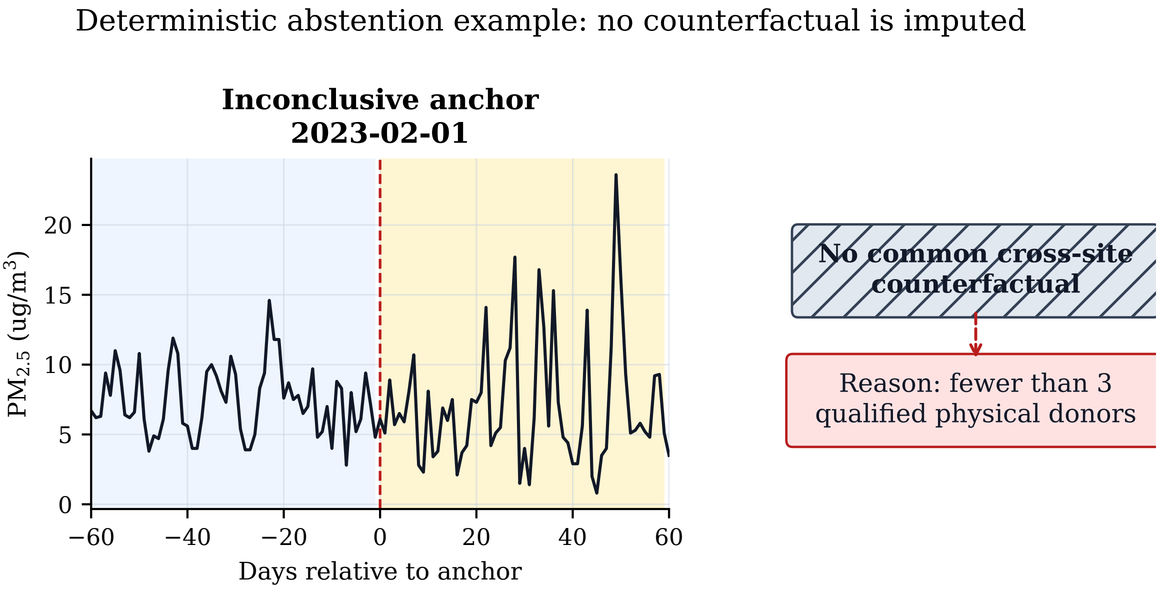

Figure 14b: Deterministic representative audit abstention. The retained target series is shown without a constructed counterfactual because fewer than three qualified distinct physical donors were available. The protocol therefore records an unavailable common comparison and does not impute a placebo, effect, or interval.

9 Discussion

MetaShift-Bench turns a weakly labelled monitoring question into an auditable computer-science benchmark. The key contribution is not an assertion that a new weighting method detects more instrument problems than existing methods. Rather, it is an evidence architecture that makes the input identity, time-respecting computation, known-truth evaluation, full event population, and failure paths inspectable together. This matters for public monitoring data because a compelling graph can conceal a non-comparable donor, an unrecorded selection decision, or an unavailable input window. A useful benchmark should make those contingencies visible rather than treating them as implementation details.

The physical-site/POC distinction is a concrete example. An AQS POC identifies a reported occurrence in the data system, not a universally stable physical instrument identity. If multiple POCs from one physical site are counted as independent donors, a target can appear to have multiple controls while its geographic evidence is not genuinely independent. The v0.3.2 donor rule therefore chooses at most one POC per physical donor site by a deterministic pre-event ranking. This choice reduces nominal availability, but it makes the meaning of a "three-donor" comparison honest. The many donor-insufficient anchors are not a defect to hide: they quantify where a cross-site method is not supported by the available data under the chosen independence proxy.

The stable-regime synthetic experiment supplies the one part of the project with known injected truth. Its target-only and matched regional perturbations directly test the computational distinction that a raw change-point does not: is a pattern specific to the target relative to comparable donors, or shared with them? The full method table makes clear why a single headline metric would be inadequate. Cross-site procedures can exploit reference information, while single-series procedures remain useful guards against overclaiming because they show what can be detected without a regional comparison. The perturbation-family matrix also avoids presenting a favorable aggregate as evidence that one method works equally well for additive, proportional, temporary, drifting, and variance-only patterns.

The result for MetaShift is deliberately negative. Cross-validated MetaShift has a favorable point estimate for local-effect MAE, but the paired event-cluster bootstrap interval against standard synthetic control crosses zero. Standard synthetic control also leads the frozen aggregate in attribution ranking and macro-F1. It would be scientifically inappropriate to convert one point estimate, selected experiment, or visually appealing case into a broad algorithm-superiority statement. The report preserves this result in the abstract, results, conclusion, source validator, and claim ledger. It also avoids further optimization on the already viewed held-out partition. A future model proposal would require a new physical-footprint-disjoint blind manifest, frozen parameters, and a precommitted metric set before outcomes are viewed.

The real-anchor audit has a different role from the synthetic benchmark. It does not validate a model against physical measurement-bias truth, which is unavailable from the Method Code field. Instead, it answers a bounded operational question: for which reported transitions can the frozen counterfactual and diagnostic workflow be executed, and what does each available diagnostic say? Complete comparisons, donor insufficiency, and input-window failures are all part of the result. This complete accounting guards against survivorship bias from reporting only well-matched stations. It also separates a necessary computational abstention from a substantive claim about the monitoring site.

The evidence tiers implement the same separation at a finer level. A supported candidate has passed a conjunction of predeclared observational diagnostics, including pre-fit quality, a conditional fixed interval, time-placebo screens, and leave-one-donor-out stability. It is not a supervised label and it does not prove a physical fault. A not-supported event has the required evidence but fails at least one diagnostic; an inconclusive event lacks a required comparative or diagnostic input. This three-way output is more informative than an all-or-nothing detector because it shows whether a low-confidence result comes from contradictory diagnostics or from unavailable data. It is also a safer decision interface for follow-up work: only humans with station records can decide whether a candidate merits an equipment-level hypothesis.

The interval results are another reason to avoid overly strong language. Fixed-weight moving-block intervals under-cover the known synthetic effects at their nominal level, while split-conformal intervals over-cover in the same frozen design. Neither observation can be corrected by post hoc adjustment without invalidating the held-out evaluation. The nested real-event procedure does capture some uncertainty from time resampling, donor re-selection, and weight refitting inside the observed candidate pool, but full selection-aware synthetic coverage at that scale was frozen as infeasible within the deadline. Thus real-event intervals are useful protocol-defined diagnostics, not calibrated confidence intervals for measurement bias. This distinction is important because numerical intervals can otherwise imply more physical certainty than their construction supports.

External context adds perspective but not ground truth. Same-site POC and hourly comparisons may reveal directional consistency, yet POCs are not universal equipment identities and temporal agreement alone cannot establish mechanism. The targeted public-document review likewise records an evidence shortfall rather than interpreting an unlocated notice as proof that no change occurred. The separate 88502 pipeline demonstrates that the machinery can be applied without pooling parameter codes, but its sparse complete-comparison population limits any transfer conclusion. These results support a workflow in which automated evidence synthesis guides a records search rather than replaces it.

Finally, reproducibility is part of the scientific result. The frozen release binds source artifact hashes, input-footprint checks, document checks, claim-to-evidence mappings, and independent environment hashes. The presentation layer adds deterministic figures and checksum-pinned case reconstruction without modifying the analysis artifacts. These safeguards cannot establish biological, engineering, or causal truth. They do make it possible for another investigator to distinguish a reproducible computation from an unsupported interpretation, which is the appropriate achievement for a metadata-anchored audit benchmark.

10 Limitations and threats to validity

The benchmark is intentionally designed to expose limitations rather than to hide them behind a binary classifier. Table 15 organizes the residual risks by construct, internal, statistical, and external validity. The mitigations constrain interpretation, but none turns the audit into a physical instrument diagnosis.

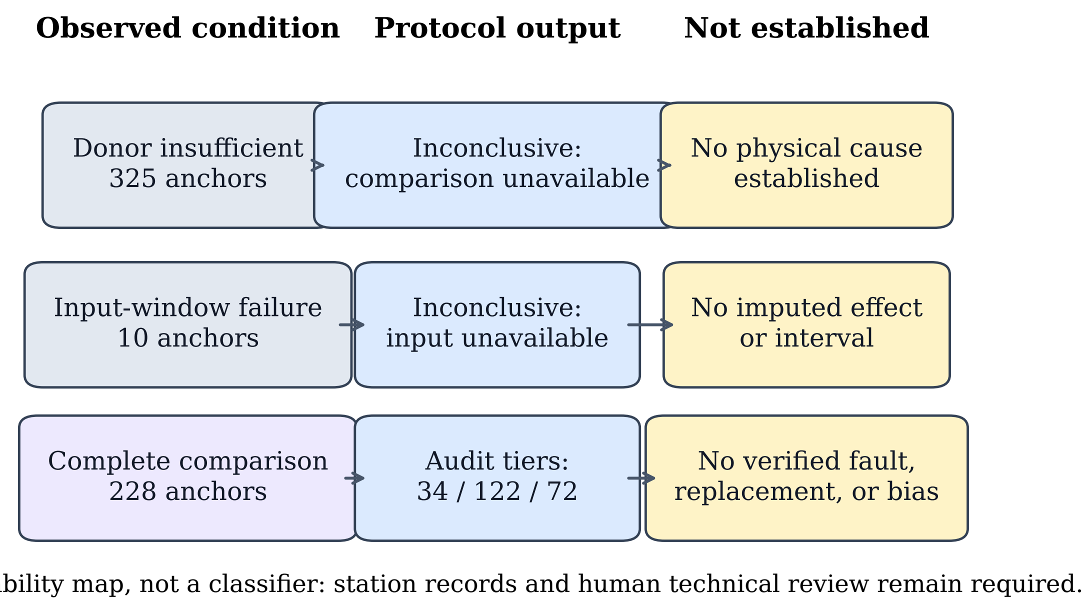

Figure 15: Applicability and failure-mode map for the frozen audit. The diagram is not a classifier: an available cross-site diagnostic result and an inconclusive availability failure both leave physical mechanism unresolved. Station histories, technical records, and human review remain necessary for any equipment-level hypothesis.

Table 15: Threats to validity, mitigation, residual consequence, and needed future evidence.

Validity class	Risk and current mitigation	Residual consequence	Needed future evidence
Construct	**Risk.** A Method Code or POC may not encode a physical device, calibration, siting, or maintenance event. **Mitigation.** Treat records as metadata anchors; prohibit fault, replacement, bias, and correction claims.	Residual effects cannot identify a physical mechanism.	Dated station histories, calibration records, deployment logs, and independently reviewed taxonomy.
Internal	**Risk.** Weather, smoke, emissions, processing, or donor instability can create a target-reference residual. **Mitigation.** Pre-event matching, distinct physical donors, time and donor placebos, and leave-one-donor-out checks.	Diagnostics can reject some explanations but do not establish randomized causal attribution.	Meteorological covariates, emissions records, network operations logs, and prospective comparator studies.
Statistical	**Risk.** Donor selection, serial dependence, multiplicity, and incomplete resampling can invalidate naive uncertainty. **Mitigation.** Fixed conditional and nested intervals, moving blocks, exploratory BH screening, and reported coverage failures.	Real-event intervals are diagnostic rather than coverage-calibrated physical-bias confidence intervals.	A computationally feasible, predeclared full selection-aware synthetic coverage study on a new blind set.
External	**Risk.** The 2019–2025 U.S. primary parameter, availability rules, stable perturbations, and donor radius may not transfer. **Mitigation.** Separate 88502 pipeline, complete event accounting, and parameter-specific reporting.	Results do not establish performance for other pollutants, countries, networks, eras, or perturbation mechanisms.	Independent publicly reproducible benchmarks with precommitted footprint-disjoint evaluation.

10.1 Construct validity

The primary construct is a reported metadata transition paired with a protocol-defined local residual, not a physical monitoring event. Method names may contain useful contextual descriptions, but they do not substitute for technical records. The benchmark therefore cannot determine whether a change reflects hardware, configuration, quality assurance, a reporting convention, or another process. The human-blocked taxonomy remains excluded from all stratified results because a row-by-row interpretation has not been completed by students and supervising teacher.

10.2 Internal validity

Cross-site counterfactuals require an assumption that qualified donors offer a useful pre-event reference. Physical-site deduplication, correlation, distance, method stability, and paired-data checks reduce obvious violations, but cannot ensure that the target and donor network remain comparable after an anchor. Regional events may be imperfectly shared, and a local residual may have non-instrument causes. Placebos and sensitivity analyses therefore constrain interpretation; they do not identify a unique mechanism.

10.3 Statistical validity

The fixed-weight interval omits uncertainty from donor selection and model selection by construction. The nested interval expands that scope inside a fixed observed candidate pool, yet the evaluation does not include a full selection-aware synthetic coverage result. The observed undercoverage and overcoverage are negative results, not defects repaired by changing confidence levels or block lengths after inspection. Multiple-screening quantities remain exploratory and do not transform all audit rows into formal causal hypothesis tests.

The implementation also centers residuals with the inclusive $t_0 - 180$ through $t_0 - 15$ calibration slice while displaying a $t_0 - 60$ through $t_0 - 1$ pre-anchor comparison, producing a 46-calendar-date overlap. This does not leak post-anchor outcomes into fitting, but it prevents the displayed pre-anchor residuals from serving as an independent validation sample. The frozen evidence is reported as implemented; no post-hoc non-overlap result is substituted for it.

10.4 External validity

The study's archive years, PM2.5 parameter, daily summary, U.S. network, and synthetic perturbation generators limit generalization. The benchmark also requires sufficiently dense, nearby, stable, and correlated donor records; its applicability will be lower in sparse networks. A separate 88502 pipeline demonstrates only limited transfer feasibility, not a broad replication. Future work should pre-register independent sources, station footprints, metrics, and external documentation before testing any newly proposed algorithmic claim.

11 Reproducibility, integrity, and contributions

11.1 Frozen evidence and computational handoff

All reported numerical evidence remains bound to the v0.3.2-evidence-final release and commit `57d678ecabebff724d898abe626c9ef80538775b`. The tracked evidence summary names every required scientific artifact, records its SHA-256, and specifies the `stable_full_v2` case-manifest checksum. The paper asset generator refuses a missing, changed, non-v0.3.2, or

Table 16: Frozen evidence release verification record.

Record	Value
Evidence tag	v0.3.2-evidence-final
Frozen evidence commit	`57d678ecabeb`
Synthetic result label	stable_full_v2
Case manifest SHA-256	`065b1b65c231c529...`
Release-gate checks passed	35/35
Document-consistency checks passed	12/12
Markdown number checks passed	57/57
Two-environment core hashes matched	34

Note. The two-environment statement applies only to the listed designated core artifacts in the frozen reproducibility comparison, not to every local file.

non-`stable_full_v2` source. It writes every displayed numeric table, vector figure, evidence macro, case-study manifest, and asset checksum into a paper-local manifest. The generator does not run the scientific pipeline or alter the frozen analysis artifacts.

Three presentation-only contracts additionally pin the deterministic stable-window illustration, the implemented inclusive window boundaries, and the QA-collocation evidence ladder. Each names its input SHA-256 values and fails if a display source changes. They make limitations visible in the report without adding any result to, or changing any value in, the frozen evidence release.

The display-only representative-case renderer adds a second provenance layer. It checks the case-selection configuration, geographic-donor table, and public archive source manifest before locally reconstructing the two complete cases. It then compares each recomputed fixed-weight MetaShift effect with the reliability-prior penalty against the saved frozen method result and fails on an absolute mismatch above 10^{-9}. This allows visual inspection of the underlying time series without committing raw EPA archives, raw API responses, credentials, virtual environments, or cache directories.

11.2 Build and review procedure

The formal-paper build invokes, in order, frozen-asset generation, claim-ledger validation, source-structure validation, reference validation, asset determinism validation, a clean LaTeX build, a font audit, and page-by-page rendering for visual review. Generated assets are deterministic; timestamps are confined to validation records rather than used as scientific inputs. The final build record reports the exact source commit, PDF SHA-256, page count, overfull-box result, font report, and visual-review result.

Reproduction claims are deliberately scoped. The two-environment comparison means that the 34 designated core artifacts listed in the frozen comparison matched by SHA-256 under the locked requirements file. It does not assert that every local file, renderer output, timestamp, or untracked raw archive is byte-identical across all systems.

11.3 Student contribution and integrity boundary

This report is a technical draft, not a competition submission. Identity, authorship, advisor relationship, student understanding, AI-use disclosure, taxonomy review, signatures, stamps, plagiarism review, and final attestation remain human-only. Required human steps are in `HUMAN_COMPLETION_CHECKLIST.md`; automation cannot complete them or create a taxonomy-stratified claim.

12 Conclusion

MetaShift-Bench is a reproducible protocol for auditing reported PM2.5 measurement-method transitions with metadata anchors, distinct physical donor controls, stable-window synthetic truth, and complete real-event accounting. Its evidence supports a benchmark-and-audit contribution: it records where cross-site diagnostics are available, where they are not, and what the available diagnostics can and cannot say.

The benchmark deliberately keeps target-specific synthetic perturbations separate from matched regional perturbations, and it compares cross-site and single-series procedures in the same frozen matrix. This design makes a reference-series contribution, a common regional pattern, and an unavailable counterfactual distinguishable outputs rather than interchangeable evidence. Its explicit donor-insufficient and input-window dispositions preserve the audit coverage boundary instead of silently converting missing comparisons into negative results. The held-out interval findings likewise state what the available constructions do not calibrate.

The frozen results do not establish a confidence-supported aggregate MetaShift advantage over standard synthetic control. They also do not establish that a reported Method Code transition represents a physical instrument change or a measurement bias. The appropriate use is therefore to prioritize a limited set of candidate station-specific discontinuities for additional records-based and technical review, while preserving unsupported and inconclusive anchors in the public audit trail. The reproducibility record in Section 11 scopes the two-environment claim to its designated core artifacts and preserves the human-only work required before a competition submission.

The v0.3.2 evidence release makes this boundary reproducible by binding source artifacts, physical-input-footprint checks, generated result tables, and claim-to-evidence mappings. Future algorithmic work can use this audit architecture only with a new, physically footprint-disjoint blind manifest and precommitted protocol; it cannot reuse the viewed evaluation partition to restate a superiority claim. This is the intended contribution: a transparent benchmark for disciplined follow-up, not a substitute for station records or human technical judgment.

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A Supplementary protocol details

A.1 Anchor-level audit pseudocode

For each primary-parameter anchor $a = (s, p, t_0, m^-, m^+)$, the frozen workflow executes the following pseudocode:

1. validate canonical daily observations and identify a persistent reported Method Code transition;
2. construct a geographic candidate pool at distinct physical sites, retaining at most one POC per site by the pre-event ranking rule;
3. record an explicit donor-insufficient disposition if fewer than three qualified physical donors exist;
4. fit cross-site weights and residual center only on the inclusive $t_0 - 180$ through $t_0 - 15$ calibration slice;
5. require at least two available donors and at least 30 retained observations in each calibration, pre-anchor, and post-anchor residual window;
6. record an explicit input-window failure if those common conditions do not hold;
7. for complete cases, compute fixed and nested diagnostic intervals, placebo diagnostics, donor sensitivity, and the rule-based evidence tier;
8. preserve every row, including all failures and unavailable diagnostics, in the audit tables.

This sequence is an implementation specification for the frozen audit. It does not claim that any step confirms a physical mechanism.

A.2 Case-study reconstruction contract

The representative-case configuration is a display contract rather than a new experiment. It fixes selection rules before rendering, binds public source archives by their saved SHA-256 values, uses the frozen data-cleaning and reliability-prior weight rule, and records the selected donor physical sites and reconstruction tolerance. If a required archive is unavailable or mismatched, the renderer fails rather than silently substituting a current public download. No raw archive is distributed with this paper.

A.3 Full perturbation-family matrix

Table 17 contains every frozen method, metric, and paired local/regional perturbation family. It is placed in the appendix for readability; Figure 7 gives a deliberately simplified main-text comparison while preserving the complete matrix here.

Table 17: Perturbation-family-specific held-out synthetic metrics for all methods.

Paired perturbation family	Method	MAE	AUPRC	Macro-F1	Regional FPR
Additive step	Standard SC	0.07788	0.99946	0.93092	0.138
Additive step	MetaShift fixed	0.08373	0.99695	0.87969	0.233
Additive step	MetaShift CV	0.07680	0.99801	0.90161	0.193
Additive step	Nearest-neighbor DiD	0.08680	0.99210	0.93614	0.105
Additive step	Bayesian mean shift	0.22860	0.50497	0.41031	0.890
Additive step	Before-after median	0.22774	0.50497	0.44778	0.808
Additive step	CUSUM	N/A	0.50497	0.43785	0.833
Additive step	Rolling MAD	N/A	0.50497	0.49862	0.553
Additive step	PELT	N/A	0.50497	0.42673	0.858
Proportional step	Standard SC	0.07635	0.85856	0.76319	0.188
Proportional step	MetaShift fixed	0.08016	0.84617	0.78293	0.090
Proportional step	MetaShift CV	0.07457	0.84296	0.78357	0.103
Proportional step	Nearest-neighbor DiD	0.08287	0.82011	0.70938	0.065
Proportional step	Bayesian mean shift	0.23444	0.50497	0.48849	0.350
Proportional step	Before-after median	0.22774	0.50497	0.46944	0.260
Proportional step	CUSUM	N/A	0.50497	0.48467	0.328
Proportional step	Rolling MAD	N/A	0.50497	0.40894	0.108
Proportional step	PELT	N/A	0.50497	0.48326	0.320
Gradual drift	Standard SC	0.08321	0.99710	0.93985	0.110
Gradual drift	MetaShift fixed	0.08551	0.99224	0.89419	0.193
Gradual drift	MetaShift CV	0.07958	0.99521	0.92082	0.153
Gradual drift	Nearest-neighbor DiD	0.08746	0.98570	0.93248	0.085
Gradual drift	Bayesian mean shift	0.24325	0.50497	0.45055	0.800
Gradual drift	Before-after median	0.23075	0.50497	0.47143	0.733
Gradual drift	CUSUM	N/A	0.50497	0.45322	0.793
Gradual drift	Rolling MAD	N/A	0.50497	0.49969	0.475
Gradual drift	PELT	N/A	0.50497	0.49992	0.513
Temporary step	Standard SC	0.16187	0.98113	0.91246	0.108
Temporary step	MetaShift fixed	0.16436	0.96551	0.88871	0.130
Temporary step	MetaShift CV	0.15875	0.96994	0.90372	0.113
Temporary step	Nearest-neighbor DiD	0.18885	0.93733	0.84114	0.065
Temporary step	Bayesian mean shift	0.28058	0.50497	0.49930	0.463
Temporary step	Before-after median	0.32696	0.50497	0.49614	0.588
Temporary step	CUSUM	N/A	0.50497	0.48686	0.660
Temporary step	Rolling MAD	N/A	0.50497	0.45747	0.780

Note. Each family aggregates its target-only local perturbation and its matched target-and-donor regional variant. Consequently, regional FPR is assessed within the same paired family; the rows are not local-only effect estimates. N/A has the same meaning as in Table 6.

Table 17, continued

Paired perturbation family	Method	MAE	AUPRC	Macro-F1	Regional FPR
Temporary step	PELT	N/A	0.50497	0.47750	0.708
Variance increase	Standard SC	N/A	0.58680	0.48825	0.158
Variance increase	MetaShift fixed	N/A	0.57484	0.45372	0.105
Variance increase	MetaShift CV	N/A	0.57314	0.46902	0.113
Variance increase	Nearest-neighbor DiD	N/A	0.56157	0.42775	0.055
Variance increase	Bayesian mean shift	N/A	0.50056	0.45595	0.205
Variance increase	Before-after median	N/A	0.50741	0.42520	0.135
Variance increase	CUSUM	N/A	0.50470	0.48012	0.290
Variance increase	Rolling MAD	N/A	0.50191	0.39371	0.093
Variance increase	PELT	N/A	0.49172	0.45495	0.213

Note. Each family aggregates its target-only local perturbation and its matched target-and-donor regional variant. Consequently, regional FPR is assessed within the same paired family; the rows are not local-only effect estimates. N/A has the same meaning as in Table 6.

Table 18: Appendix-only temporal concentration of reported 88101 metadata anchors.

Old code	New code	Abbreviated reported metadata change	Count
236	636	T640, baseline reported name -> T640, alignment-enabled	234
238	638	T640X, baseline reported name -> T640X, alignment-enabled	120

Note. Of 393 2023 anchors, these two pairs account for 354 (90.1%). The largest date is 2023-07-01 (50), and the largest state-code subtotal is 42 (42). Descriptions abbreviate reported AQS Method Name strings; they do not establish a cause.

A.4 Descriptive concentration of reported anchors in 2023

The frozen inventory has 393 of 563 anchors (69.8%) dated in 2023. Two named reported code-pair transitions, 236 to 636 and 238 to 638, account for 354 of those 393 anchors (90.1%). Their new Method Name strings include "Network Data Alignment enabled." Figure 16 and Table 18 make that concentration visible because it is relevant to interpretation and possible future records requests.

This pattern is descriptive only. A code-pair label, a temporal cluster, or an EPA method-context document does not establish an administrative, software, hardware, policy, calibration, or data-processing cause for any site-level record. The human-blocked taxonomy review remains outside this report's stratified analysis.

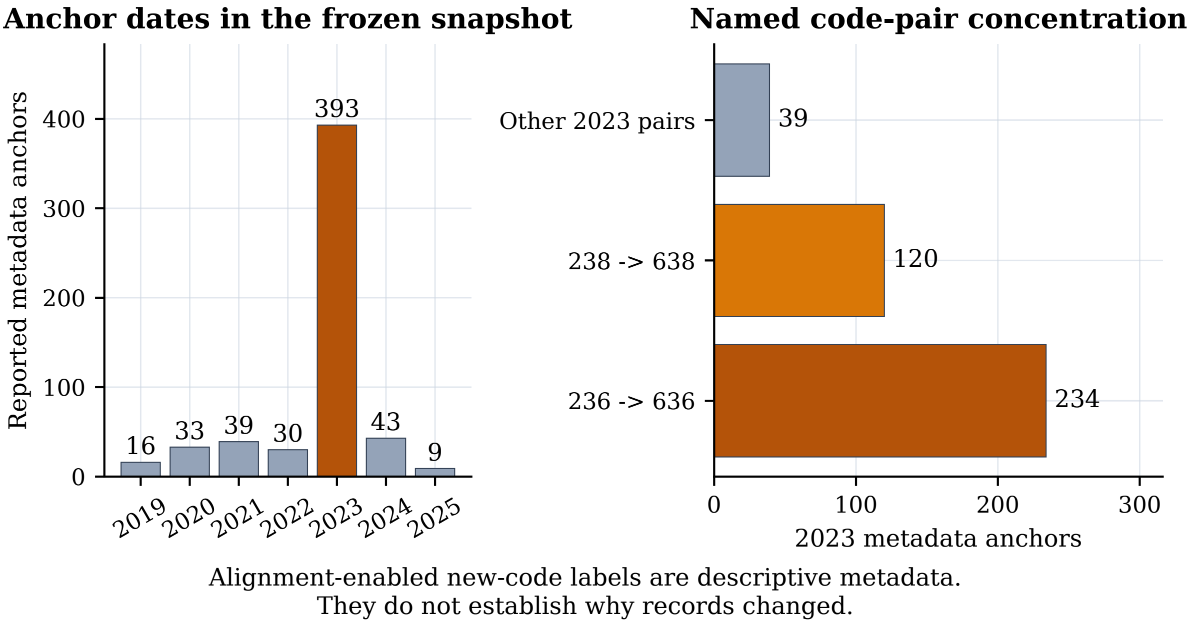

Figure 16: Appendix-only descriptive concentration of reported 88101 metadata anchors. The left panel shows the frozen date distribution; the right panel shows the two named code-pair counts within 2023. The association between labels and dates is not evidence of a physical or administrative cause.

B Acknowledgements, contribution statement, and required disclosures

HUMAN COMPLETION REQUIRED. This section is an intentionally uncompleted truthful-disclosure template. It must be replaced by a 500–1500 word account, verified by every student author and supervising teacher, before the report is submitted. Do not state that a person performed work they did not perform. Do not state that an advisor, school, organization, or AI system did or did not contribute unless the authors have verified the statement. Before submission, all student authors and supervising teachers must verify the exact final PDF and complete the required academic-integrity declaration, signatures, institutional stamps, and final truthfulness attestation.

Topic origin

HUMAN COMPLETION REQUIRED: Explain how the team selected the question, including the intellectual path from the original observation or reading to the metadata-anchor audit formulation. State whether a supervising teacher or other person suggested the topic and how the students refined the question.

Data acquisition and analysis

HUMAN COMPLETION REQUIRED: Identify who downloaded or reconstructed the public EPA AirData inputs, who inspected data rules, who implemented or reviewed the cleaning and donor-uniqueness logic, and who ran or checked the frozen analyses. State only the actual use of public AQS bulk files, public API endpoints, and local credentials. Do not include credentials, raw API responses, or private information.

Experimental design and writing

HUMAN COMPLETION REQUIRED: State which student(s) designed the synthetic evaluation, maintained the no-tuning boundary, generated figures and tables, interpreted limitations, and drafted or revised each report section. Describe the physical-donor uniqueness defect and remediation only if the authors personally understand and verify that account.

Advisor relationship, compensation, and external help

HUMAN COMPLETION REQUIRED: List each supervising teacher or advisor, their institution and permitted role, the nature of their guidance, and whether any paid tutoring, commercial training, external code, data, or editorial help was involved. Competition rules prohibit untruthful disclosure and may restrict commercial guidance; use the official form for final confirmation.

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